

RAY-VASC: Residual Adaptive Capacity, Neuropathic Repair-Remap, and Adaptive Neurotrophic Research Architecture in Complex Neurovascular Neurodegeneration

Canonical Thesis | PhD Submission Edition

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Originating case: RAY-001 (de-identified case identifier)

Abstract

Background. Complex neurovascular neurodegeneration creates a measurement problem as much as a treatment problem. Progressive supranuclear palsy, vascular cognitive impairment, post-stroke network disruption, traumatic brain injury burden, sleep fragmentation, respiratory limitation, and metabolic instability can coexist in a single phenotype while producing overlapping cognitive, motor, affective, and autonomic signals. Conventional follow-up intervals and single-domain outcomes often cannot distinguish irreversible structural progression from reversible physiological suppression. RAY-VASC addresses this gap by proposing a longitudinal research architecture for testing residual adaptive capacity in surviving neural systems.

Objective. This thesis formalizes the Neuropathic Repair-Remap Hypothesis (NRRH), which proposes that complex neurovascular phenotypes may retain measurable adaptive capacity within surviving neural systems. It evaluates whether substrate priming, multimodal terrain stabilization, a conditionally governed plasticity window, and transparent response-policy rules can improve the interpretability of longitudinal functional trajectories. The thesis tests plasticity readiness and network accessibility; it does not infer therapeutic efficacy from architecture alone.

Design. The proposed architecture is a prospective, longitudinal N-of-1 and small-series research program with preregistered outcomes, evidence-graded candidate inputs, safety filters, confounder annotation, and formal falsification criteria.

Case. RAY-001 is the de-identified originating case, characterized by PSP, vascular cognitive impairment/dementia, prior infarcts, abnormal DaTSCAN, COPD, severe sleep fragmentation, polypharmacy, falls, dysphagia, and traumatic exposure history. Its complexity defines the baseline, safety, and measurement requirements used to test the architecture.

Architecture. The candidate universe is separated from participant-specific eligibility, and APPRE operationalizes exposure tracking, confounder capture, safety review, interpretability grading, and prospective policy updates.

Primary Outcomes. Feasibility, adherence, safety, signal interpretability, functional trajectory, EEG dynamics, sleep metrics, inflammatory burden, neurotrophic markers, and biomarker trajectories form the core evaluation spine.

Significance. RAY-VASC contributes a falsifiable, measurement-first framework for studying residual adaptive capacity in complex neurovascular neurodegeneration and for translating high-uncertainty mechanistic hypotheses into auditable research designs.

Acknowledgments

This thesis originates in a family crisis that forced a translational research question into practical form. RAY-001 is acknowledged as the originating case whose complexity made the architecture necessary. Caregiver observation, clinical documentation, and the published RAY-VASC manuscript series created the source base from which this thesis was built. This acknowledgment records provenance: a real-world phenotype exposed a measurement gap, and the research program was designed to make that gap testable.

Author Declaration and Contributions

Logan Henderson is the thesis author, program architect, and principal developer of the RAY-VASC research architecture, including the Neuropathic Repair-Remap Hypothesis, the Regenerative Loop, the Three-Phase Architecture, and the APPRE schema-and-rules framework.

Public case references use the de-identified identifier RAY-001. Legal-name use is authorization-dependent and is not required for the scientific argument. Internal project documents are used for case history and program architecture; peer-reviewed and official sources are used for scientific claims. Verification-pending literature leads are quarantined in Appendix K and are not used as load-bearing evidence. This thesis is not medical advice, legal advice, or a clinical directive. It is for research planning, academic documentation, open-science governance, and dissertation scaffolding only. All health decisions must remain under licensed clinician oversight. Controlled-substance content is limited to published evidence and lawful, clinician-governed, ethics-approved research contexts.

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List of Abbreviations and Key Terms

Term	Definition in this thesis
APPRE	Algorithmic Personalization Predictive Response Engine: a transparent schema-and-rules framework for response-policy documentation, confounder annotation, and interpretability grading.
BDNF	Brain-derived neurotrophic factor; used as a plasticity-relevant biomarker and mechanistic pathway marker, not as a standalone clinical proof.
CENT	CONSORT Extension for Reporting N-of-1 Trials: a reporting standard used to structure N-of-1 transparency.
DaTSCAN	Dopamine transporter imaging is used clinically to support dopaminergic degeneration interpretation, not disease-specific proof by itself.
GPW	Governed Plasticity Window: conditional, lawful, clinician-governed, ethics-approved research window.
NRRH	Neuropathic Repair-Remap Hypothesis: the thesis claims that residual adaptive capacity in surviving systems can be studied through staged terrain stabilization, plasticity-readiness measurement, and response-policy logic.
PLMI	Periodic Limb Movement Index: a sleep-fragmentation metric with major confounder importance in RAY-001.
PSP	Progressive supranuclear palsy: a progressive 4-repeat tauopathy and major disease-context anchor.
RAY-001	Default case identifier for the originating stress-test case.
Regenerative Loop	The terrain framework linking sleep, exertion, light/circadian rhythm, respiration, nutrition/hydration, social engagement, and safety.
Responseome	A structured longitudinal map of how participant domains move in relation to exposures, confounders, terrain variables, safety events, and policy decisions.
Three-Phase Architecture	Pre-Load, Governed Plasticity Window, and Sustain and Integration.
VCID	Vascular cognitive impairment and dementia.

Reader Orientation: Published RAY-VASC Manuscript Series, Source Hierarchy, and Evidence Grades

This section identifies the published manuscript series, source hierarchy, evidence-grading rules, and chapter functions used throughout the thesis.

Table 1. Published RAY-VASC manuscript series.

Paper	Public title	Thesis role	Primary DOI / public record
Paper 1	<i>The Brain That Remains</i>	Residual adaptive capacity and open-source protocol logic	10.5281/zenodo.21109062
Paper 2	<i>RAY-VASC Protocol</i>	N-of-1/small-series measurement architecture	10.5281/zenodo.21111619; OSF 10.17605/OSF.IO/WESZY
Paper 3	<i>RAY-001 Baseline Characterization</i>	Originating case and baseline feasibility foundation	10.5281/zenodo.21113371
Paper 4	<i>APPRE Schema-and-Rules Framework</i>	Computational/schema foundation	10.5281/zenodo.21116291

Table 2. Project source index and evidence hierarchy.

Source class	Thesis role	Primary contribution	Boundary
Established citable RAY-VASC Papers 1-4	Published RAY-VASC manuscript series	Establish theory, protocol, baseline case characterization, and APPRE schema	Citable program documents; not independent clinical efficacy evidence
Internal RAY-VASC source files	Case and architecture provenance	Preserve clinical source synthesis, stack logic, APPRE/LIME design, and longitudinal forms	Used for program history and de-identified case facts; not peer-reviewed evidence
External peer-reviewed literature	Scientific evidence-based	PSP, VCID, TBI, neuroplasticity, Bacopa, mycology, psilocybin, biomarkers, N-of-1 methods	Supports scientific claims according to design, population match, and evidence grade
Official sources and reporting standards	Governance and methods guardrails	CENT, CONSORT, SPIRIT/SPENT, regulatory and ethics boundaries where applicable	Used for methods discipline and de-identified documentation boundary, not biological proof
Verification-pending literature leads	Research tracking only	Preserve emerging claims that may become relevant after verification	Quarantined in Appendix K; not load-bearing

Table 3. Evidence grading legend used throughout the thesis.

Grade	Definition	Permitted thesis use
A	Multiple RCTs, meta-analyses, regulated Phase 3 data, or high-confidence reporting standards	May support strong general claims only when the population and endpoint match
B	Some RCTs, Phase 1/2 human studies, validated instruments, or strong observational evidence	Supports moderate claims and method choices
C	Case report, pilot, small-N study, feasibility study, or internal baseline characterization	Hypothesis-generating; cannot establish causality
D	Animal, in vitro, or preclinical mechanistic evidence	Mechanistic plausibility only; not clinical proof
E	Theory, traditional use, expert framing, project hypothesis, or verification-pending claim	Label explicitly; do not treat as fact

Table 4. Thesis architecture and chapter function.

Chapter	Title	Distinct job	Primary formal object
I	Introduction and thesis statement	Define the measurement void, NRRH, contribution, scope, and falsifiability	Thesis statement and published-paper relationship
II	Case characterization: RAY-001	Orient the reader to the originating stress-test phenotype	Diagnostic, risk, baseline, medication, and confounder tables

Chapter	Title	Distinct job	Primary formal object
III	Literature review	Establish the disease, neuroplasticity, Layer 1 candidate-universe, biomarker, and methods, evidence-based	Evidence-graded literature synthesis
IV	Theoretical framework	Formalize NRRH, Regenerative Loop, substrate priming, and Three-Phase Architecture	Mechanism-to-measurement crosswalk
V	Computational framework: APPRE	Define the response-policy schema, feedback loop, governance gates, and interpretability logic	APPRE schema and rules tables
VI	Methodology	Specify design, outcomes, safety, confounder capture, analysis, and reproducibility	Implementable protocol spine
VII	Baseline findings and preliminary dataset	Separate the confirmed baseline from future outcomes	Baseline/gap/feasibility tables
VIII	Discussion	Interpret the contribution, limitations, falsification, and future research path	Field contribution and next-study rationale
IX	Conclusion	Restate the thesis and define a collaborative path forward	Academic synthesis and research call
Appendices A-L	Conventional institutional end matter	Preserve full tables, data dictionary, safety matrices, source integrity, and reporting checklist	Audit-ready institutional end matter

Chapter I - Introduction and Thesis Statement

1.1 The therapeutic and measurement void in complex neurovascular neurodegeneration

Complex neurovascular neurodegeneration does not manifest as a single disease process with a singular outcome. Instead, it presents as an interactive system where tauopathy, cerebrovascular injury, sleep fragmentation, respiratory limitations, medication effects, metabolic instability, recurrent falls, and caregiver assessments all influence the observable phenotype. A participant may appear to decline due to various factors: progression of structural injury, increase in reversible terrain burden, deterioration in sleep quality, worsening respiratory status, changes in medications, or a combination of these variables. Standard clinical follow-ups are crucial, but the typical three-to-six-month interval is poorly suited for differentiating these processes, particularly since they can fluctuate daily, weekly, or monthly.

Progressive supranuclear palsy (PSP) and vascular cognitive impairment exemplify this issue. PSP is a progressive tauopathy characterized by motor, ocular, postural, speech, and cognitive features, and it currently has no established disease-modifying therapy in routine clinical care (Hoglinger et al., 2017; Armstrong, 2018) [Grade A/B]. Vascular cognitive impairment results from cerebrovascular injury, small-vessel disease, infarction, disconnection, and mixed pathology, similarly lacking a single disease-modifying pharmacological solution (Gorelick et al., 2011; Dichgans & Leys, 2017) [Grade A]. When these pathologies coexist alongside sleep and respiratory disruptions, interpreting clinical signals becomes increasingly challenging. The central issue lies not only in the absence of curative therapies but also in the lack of a high-frequency framework to measure whether residual function can be stabilized, enhanced, or differentiated from confounder-driven suppression.

RAY-VASC seeks to address this measurement gap by investigating whether surviving neural circuits, peri-lesional networks, and partially suppressed functional systems can be characterized with a level of detail not achievable through conventional follow-up. The primary question of research is whether a staged, multidomain framework can make residual adaptive capacity measurable in complex neurovascular neurodegeneration.

Neurological care often condenses fluctuating states into short examinations and caregiver reports. A mixed neurovascular phenotype may vary depending on the time of day, medication timing, sleep quality, respiratory stability, and recent acute events; thus, high-frequency observation is essential to identify state transitions that conventional follow-up cannot clarify.

Consequently, RAY-VASC treats longitudinal signal detection as a primary methodological concern: differentiating structural injury from functional accessibility, progression from terrain burden, medication effects from potential confounding influences, and the quality of observation from true change.

This framework treats feasibility as a scientific endpoint. If a complex phenotype cannot produce sufficient clean observations for interpretation, the initial conclusion is methodological rather than therapeutic. The protocol must be simplified until the data stream stabilizes. This perspective differs from most supplements or lifestyle narratives. RAY-VASC does not start by asking whether a candidate works; rather, it begins by questioning whether a candidate can be safely observed and honestly interpreted in a person whose baseline state is biologically unstable.

1.2 The Neuropathic Repair-Remap Hypothesis (NRRH) - load-bearing intellectual contribution

The Neuropathic Repair-Remap Hypothesis proposes that complex neurovascular neurodegenerative phenotypes may retain residual adaptive capacity within surviving neural systems and that this capacity can be studied when biological terrain, network state, safety constraints, and outcome trajectories are measured together. This section introduces the claim only; the full four-operation model - Reinforce, Remap, Re-link, and Prevent - is formalized in Chapter IV with mechanism-to-measurement crosswalks.

The analytical distinction is between structural loss and functional accessibility within surviving systems. This distinction guides the mechanism-to-measurement model, outcome hierarchy, and falsification criteria developed in Chapters IV-VI.

1.3 The RAY-VASC research architecture

RAY-VASC is a longitudinal N-of-1 and small-series research architecture designed to test whether residual functional capacity in complex neurovascular phenotypes can be characterized and tracked under a structured, multidomain measurement program. It integrates baseline case characterization, graded evidence mapping for candidate inputs, explicit safety and eligibility filtering, high-frequency subjective and objective measurement, and preregistered analytic rules.

The architecture separates mechanistic plausibility from implementation eligibility. Layer 1 defines the full candidate universe: functional mycology, nootropics, neuroprotective agents, micronutrients, mitochondrial supports, adaptogens, sleep and circadian interventions, vascular and metabolic supports, and governed experimental items identified in the wider literature. Layer 2 applies participant-specific constraints - medication interactions, renal function, respiratory status, dysphagia, falls risk, and caregiver burden - to determine what is implementable at a given time.

RAY-001 supplies the originating constraints and baseline dataset; the scientific object is the measurement and decision framework. The thesis, therefore, evaluates RAY-VASC as a generalizable architecture for converting biologically plausible but high-uncertainty neuroplasticity hypotheses into auditable research programs, rather than as advocacy for a single stack or participant.

The architecture also separates formal hypotheses from exploratory mechanistic intuitions. Assumptions about substrate priming, plasticity windows, or network reorganization are translated into prespecified measures, phases, safety gates, and falsification criteria rather than stronger outcome claims.

1.4 Relationship to the seven-paper program

Table 1 identifies the published RAY-VASC manuscript series. Papers 1-4 establish the theoretical framework, protocol, baseline case characterization, and APPRE schema. Papers 5-7 are planned extensions addressing the Regenerative Loop, the Neurotrophic Mycology Evidence Map, and the Three-Phase Architecture.

The thesis synthesizes these contributions into one research program and defines the evidence required to support, weaken, or reject the integrated model. The published manuscripts provide project provenance; external peer-reviewed and official sources remain the basis for scientific claims.

Terminology is fixed throughout: residual adaptive capacity denotes the measurable property under investigation, plasticity readiness denotes terrain conditions permissive of adaptive change, and network accessibility denotes recruitment of surviving functional systems.

1.5 Scope and falsifiability - what would support, weaken, or defeat the thesis

The thesis would be supported if the architecture demonstrates feasibility, safety documentation, interpretable data density, repeated within-person trajectory signals, and objective marker correspondence across at least some domains. It would be strengthened if terrain stabilization precedes functional or network improvements, if EEG or biomarker changes align with functional trajectories, and if APPRE reduces attribution error by separating clean cycles from confounded cycles.

The thesis would be weakened if adherence is insufficient for analysis, if the Layer 1 candidate universe cannot be filtered safely, if signals remain anecdotal or single-domain, if objective markers do not correspond to observed function, or if small-series extension fails to reproduce the measurement architecture. It would be defeated as a research architecture if it cannot produce interpretable cycles under realistic caregiver and clinical constraints. These falsification conditions are part of the design rather than post-hoc defenses.

Claim boundary. This thesis does not claim therapeutic efficacy for any candidate compound, protocol, stack, or delivery method, and it does not propose APPRE as a clinical decision engine or treatment recommender. The architecture is explicitly measurement-first: it tests plasticity readiness, terrain stabilization, and network accessibility under rule-bound exposure states, and documents confounders, safety events, and interpretability grades. Any observed functional or biomarker changes are treated as within-person research signals that require future controlled trials, not as proof of benefit. Clinical decisions must remain under licensed clinician oversight, and controlled-substance use is limited to lawful, ethics-approved, clinician-governed research contexts only.

Chapter II - Case Characterization: RAY-001

2.1 Phenotype overview and originating context

RAY-001 is the originating and stress-test case for the RAY-VASC architecture. The case combines progressive supranuclear palsy, vascular dementia with frontotemporal features, prior infarcts, abnormal dopaminergic imaging, severe sleep disruption, COPD, dysphagia, falls, medication complexity, and traumatic exposure history. The scientific role of the case is not to prove the thesis. Its role is to force the architecture to confront the hardest interpretation problem: a phenotype in which almost every observed change has multiple plausible causes.

Table II-1. Confirmed diagnoses and clinical evidence

Diagnosis/domain	Confirmed evidence	Primary source type	Protocol implication	Evidence grade
Progressive supranuclear palsy	Vertical gaze palsy, procerus sign, eyelid opening apraxia, dysarthria, DaTSCAN abnormality, H&Y Stage 3, MDS-UPDRS motor 52	UTMB neurology consultation	Defines motor and oculomotor baseline and high-risk phenotype	Clinical case evidence
Vascular dementia / VCID	MRI small-vessel ischemic disease, bilateral thalamic infarcts, old cerebellar infarct, MoCA 11/30	Imaging and cognitive baseline	Requires vascular-confounder tracking and cognitive outcome caution	Clinical case evidence
Primary parkinsonism	Prior diagnosis and dopaminergic treatment history	Clinical history	Creates medication interaction and attribution complexity	Clinical case evidence
Obstructive sleep apnea / severe sleep fragmentation	Sleep efficiency 39.48%; PLMI 137.66/hr; CPAP intolerance in real-world use	Sleep study and case source	Major reversible suppression candidate and confounder	Clinical case evidence
COPD / respiratory limitation	Nebulizer therapy and respiratory diagnosis	Clinical history	Requires SpO2/exertion monitoring and safety gates	Clinical case evidence
Dysphagia and falls	Soft/pureed diet, recurrent falls, fall with loss of consciousness	Clinical history	Affects feasibility, adherence, aspiration risk, and safety escalation	Clinical case evidence

2.2 Multi-hit pathological architecture and domain interaction

The case is best understood as a multi-hit system rather than a single diagnosis. PSP contributes tau-associated motor, ocular, postural, and cognitive burden. Vascular injury contributes to disconnection, executive dysfunction, gait instability, and thalamic relay disruption. Sleep fragmentation adds a reversible but severe suppressive layer. COPD and respiratory variability can alter exertion tolerance and cognition. Polypharmacy affects sleep, motor timing, blood pressure, falls, and attribution. The combination creates a measurement environment in which unstructured observation is unreliable.

Table II-2. Compounding risk factors and the pathological interaction model

Risk factor	Pathological interaction	Signal distortion risk	Measurement response
Bilateral thalamic infarcts	Relay network disruption affecting attention, sleep, motor control, and arousal	Cognitive/motor fluctuations may reflect relay instability rather than intervention response	Track sleep, motor, cognition, and EEG together
Small-vessel white-matter disease	Slow disconnection across frontal-subcortical and association networks	Executive slowing may progress independent of PSP	Use trajectory benchmarks and repeated measures
Boxing/TBI exposure	Subconcussive and concussive burden may contribute to tau/inflammatory vulnerability	Symptoms may reflect a mixed neurodegenerative history	Preserve as a non-removable confounder
COPD	Hypoxemia/exertion limitation can suppress cognition and motor performance	Daily performance may track oxygenation rather than neuroplasticity	Capture SpO2, dyspnea, and exertion context
PLMD/OSA	Sleep fragmentation impairs glymphatic clearance, attention, mood, and motor learning	False decline or false improvement if sleep changes	Capture sleep efficiency, PLMI treatment status, and daytime somnolence

Risk factor	Pathological interaction	Signal distortion risk	Measurement response
Polypharmacy	Medication timing and adverse effects alter mood, sleep, falls, and cognition	Attribution instability	Daily medication log and interaction flags

2.3 The therapeutic void in this case

The therapeutic void is concrete. Cholinesterase and NMDA-directed agents were attempted historically and were not tolerated because insomnia worsened. Dopaminergic therapy remains relevant to Parkinsonism but does not address PSP progression, vascular cognitive impairment, sleep fragmentation, or network-level residual capacity. Sleep treatment is clinically necessary but difficult because documented response during testing does not automatically translate into real-world adherence or tolerance. Rehabilitation is essential but cannot by itself resolve the attribution problem unless measured alongside sleep, respiratory, medication, and inflammatory context.

Table II-3. Medication list with interaction flags relevant to protocol

Medication/class	Relevant concern	Affected domains	Protocol implication
Sinemet IR 25/100	Dopaminergic timing alters motor and cognitive observations	Motor, mood, sleep	Anchor motor tests to medication timing
Trazodone	Sedation and sleep architecture effects	Sleep, cognition, falls	Record dose timing and morning sedation
Clopidogrel suspension	Bleeding risk with some supplements	Safety, falls, bruising	Exclude or flag antiplatelet-interacting candidates
Amlodipine/lisinopril/metoprolol	Blood pressure and heart-rate effects	Falls, exertion tolerance, fatigue	Monitor orthostasis and timing
Rosuvastatin	Muscle symptoms and metabolic context	Motor, fatigue	Record myalgia/fatigue changes
Nebulized albuterol/ipratropium	Respiratory status and stimulant-like effects	Dyspnea, HR, tremor	Log rescue use and SpO2 context

2.4 Why this case originated as the RAY-VASC case

A cleaner phenotype would be easier to study but less informative for architecture design. RAY-001 required the program to separate hypothesis, evidence, safety, confounding, implementation, and interpretation, and to treat missingness, caregiver burden, medication timing, sleep quality, respiratory status, falls, and acute events as core variables.

Table II-4. Baseline functional metrics

Metric	Baseline value/status	Interpretation	Follow-up role
MoCA	11/30	Severe cognitive impairment baseline	Repeated cognitive anchor; avoid overinterpreting single changes
MDS-UPDRS III motor	52	Substantial Parkinsonian motor burden	Motor trajectory anchor
Hoehn & Yahr	Stage 3	Postural instability with preserved some ambulation	Falls and mobility risk stratification
Sleep efficiency	39.48%	Severe sleep suppression	Regenerative Loop priority and confounder
PLMI	137.66/hr	Severe periodic limb movement burden	Sleep clinic and confounder target
A1c	5.8% historically	Prediabetes-range metabolic terrain	Update required for current analysis
B12	328 pg/mL historically	Low-normal neurocognitive cofactor	Update and interpret cautiously
eGFR	65 baseline with nadir 44	Renal variability	Safety filter for Layer 1 candidate universe

A stress-test case has methodological value because it exposes weak assumptions. A simple case can hide an underdeveloped protocol. A complex case forces the research system to decide what counts as evidence, what counts as safety exclusion, what counts as confounding, and what counts as missing information. RAY-001 required that

distinction at every layer. The presence of antiplatelet therapy alters the supplement risk map. Dopaminergic therapy changes motor attribution. Renal variability changes eligibility for candidates that might otherwise appear low-risk. Dysphagia changes feasibility and delivery assumptions. Severe PLMD and low sleep efficiency create a suppressive terrain that can dominate cognition and motor learning. COPD makes exertion and oxygenation part of the outcome system rather than background medical history.

The case chapter establishes the operational problem that the remainder of the thesis generalizes: whether severe clinical complexity can be converted into a reproducible measurement system.

The case also exposes the difference between clinical significance and research interpretability. A caregiver may correctly perceive meaningful change in speech, engagement, sleep, mobility, continence, or affect. That observation can be humanly important while remaining scientifically ambiguous. RAY-VASC does not dismiss caregiver observation. It structures it. Caregiver notes become analyzable when they are time-stamped, linked to sleep and medication context, associated with objective or semi-objective measures where feasible, and interpreted under APPRE cycle grading.

The same principle applies to negative events. A fall, infection, medication change, or respiratory exacerbation is not merely a disruption. It is data about the phenotype and the feasibility of the architecture. A protocol that cannot absorb such events is not suitable for complex neurodegeneration. The design must allow pauses, holds, downgrades, and restarts without treating every deviation as failure.

The imaging record makes this point concrete. Bilateral thalamic infarcts, small-vessel disease, generalized atrophy, old cerebellar infarct, abnormal DaTSCAN, and traumatic context create overlapping explanatory pathways. If gait improves, the mechanism could involve motor learning, sleep, dopamine timing, confidence, respiratory capacity, physical therapy, or measurement variability. If cognition worsens, the mechanism could involve tau progression, vascular burden, sleep collapse, medication effect, fatigue, or acute illness. The case requires analytic humility.

2.5 Confounder architecture and attribution challenge

The baseline dataset contains both confirmed values and missing values. A submission-grade thesis must keep those categories separate. Confirmed values anchor the existing case characterization. Needed and planned values define the next research gate. Ordered-not-yet-returned or pending values cannot be used as if confirmed. This distinction is central to APPRE because interpretability depends on knowing not only what changed but what was actually measured.

Table II-5. Lab gap analysis - confirmed vs. needed vs. planned

Domain	Confirmed	Needed	Planned role
Neurodegeneration biomarkers	Clinical diagnosis and imaging; p-tau217 not yet confirmed	p-tau217, NfL if feasible	Trajectory and differential-pathology context
Neurotrophic markers	No stable BDNF series yet	Serum/plasma BDNF methodology decision	Plasticity-readiness exploratory marker
Inflammation	WBC elevation has been noted historically	hs-CRP, IL-6, TNF-alpha if available	Inflammatory-burden tracking
Metabolic/renal	A1c/eGFR historical values	Updated CMP, A1c, lipids	Safety and terrain model
Sleep	A diagnostic sleep study exists	Current treatment status and adherence	Interpretability gate
EEG	Not yet longitudinal	Baseline and repeated resting/ERP EEG	Network trajectory marker

Table II-6. Imaging summary

Study	Date/status	Finding	Thesis relevance
MRI brain	Nov 2023	Moderate-severe chronic small-vessel disease, generalized atrophy, old bilateral thalamic infarcts, old right cerebellar infarct	Structural lesion map and VCID anchor

Study	Date/status	Finding	Thesis relevance
CT head/orbit	Aug 2025	Post-traumatic right orbital floor deformity noted	TBI history support and fall-risk context
DaTSCAN	Jan 2026	Severely reduced striatal uptake; quantitative software unable to process	Dopaminergic degeneration support
MRI brain w/wo	Feb 27, 2026, planned/pending in source spine	Pending in thesis source set	Not included as a confirmed baseline until the source update is available

Chapter III - Literature Review

3.1 Progressive supranuclear palsy: pathophysiology, diagnostic criteria, natural history, active trials, therapeutic void

Progressive supranuclear palsy is a primary tauopathy characterized by vertical gaze palsy, postural instability, axial rigidity, dysarthria, dysphagia, cognitive and behavioral change, and poor response to dopaminergic therapy relative to idiopathic Parkinson's disease. The Movement Disorder Society criteria provide the diagnostic framework for PSP syndromes and emphasize combinations of ocular motor dysfunction, postural instability, akinesia, and cognitive dysfunction (Hoglinger et al., 2017) [Grade A]. Therapeutic development has not produced a routine disease-modifying therapy. Large interventional trials such as davunetide illustrate both the feasibility of PSP trials and the difficulty of modifying progression (Boxer et al., 2014) [Grade A].

Table III-1. PSP evidence summary

Evidence domain	Key point	Representative source	Grade	RAY-VASC role
Diagnosis	MDS criteria define PSP phenotypes and diagnostic certainty	Hoglinger et al., 2017	A	Sets diagnostic language
Natural history	PSP commonly progresses faster than idiopathic PD and lacks disease modification	Armstrong, 2018; Williams & Lees, 2009	B	Trajectory benchmark
Failed disease-modifying trials	Davunetide did not meet efficacy endpoints in the PSP trial	Boxer et al., 2014	A	Shows therapeutic void
Clinical scale	PSPRS supports severity and progression tracking	Golbe & Ohman-Strickland, 2007	B	Outcome benchmark

3.2 Vascular cognitive impairment and dementia: mechanisms, bilateral thalamic infarct model, therapeutic void

Vascular cognitive impairment arises from infarction, small-vessel disease, microvascular dysfunction, hypoperfusion, white-matter injury, and mixed neurodegenerative pathology. Its clinical expression commonly involves executive dysfunction, slowed processing, gait impairment, mood change, and attentional instability. Bilateral thalamic infarcts are especially relevant because the thalamus participates in relay, arousal, memory, motor, and attentional systems. A thalamic lesion can therefore amplify functional disruption beyond the size of the infarct itself.

Table III-2. VCID evidence summary

Mechanism	Evidence summary	Representative source	Grade	Thesis role
Small-vessel disease	White-matter disease contributes to cognitive and gait impairment	Wardlaw et al., 2013	A	Lesion-network model
Vascular cognitive impairment	Vascular injury can interact with neurodegenerative pathology	Dichgans & Leys, 2017; Gorelick et al., 2011	A	Mixed-pathology rationale
Thalamic infarcts	The thalamus functions as a relay hub for cognition, arousal, emotion, and motor coordination	Clinical neuroanatomy sources: internal imaging	B/C	Relay-disconnection model

3.3 TBI as compounding accelerant: CTE risk, boxing exposure, tau pathology acceleration

Traumatic brain injury and repetitive head impacts are not treated here as a diagnostic claim of chronic traumatic encephalopathy. They are treated as a plausible compounding factor that increases uncertainty around tauopathy, inflammation, white-matter vulnerability, executive dysfunction, and falls. A mixed phenotype with PSP, VCID, and a history of boxing exposure cannot be attributed to all cognitive or motor changes to one disease process. TBI history, therefore, becomes a confounder architecture variable rather than a rhetorical explanation.

Table III-3. TBI and repetitive head-impact literature role

Topic	Evidence claim	Grade	Protocol implication
Adolescent exposure	Repetitive head impacts during neurodevelopment may increase later vulnerability	C/E depending source	TBI is treated as a non-removable confounder
CTE/tau risk	Contact-sport exposure is associated with later-life tau pathology in selected autopsy series	C	No diagnosis inferred without pathology
Falls and secondary injury	Falls add acute confounding to baseline and longitudinal data	Clinical case evidence	Event log required

3.4 Neuroplasticity mechanisms: BDNF/TrkB, NGF, synaptogenesis, glymphatic clearance, network reorganization

The thesis relies on a conservative interpretation of neuroplasticity. Plasticity does not mean reversal of neurodegeneration. It means the nervous system can alter synaptic strength, dendritic complexity, network coordination, learning capacity, and compensatory routing within biological constraints. Stroke and TBI recovery literature supports the principle that surviving networks can reorganize after injury (Cramer, 2008; Murphy & Corbett, 2009) [Grade B]. Sleep science shows that sleep state influences metabolite clearance and neural function (Iliff et al., 2012; Xie et al., 2013) [Grade B/D]. Exercise and BDNF literature support biological plausibility for activity-dependent plasticity markers (Szuhany et al., 2015) [Grade A].

Table III-4. Neuroplasticity mechanism evidence grades

Mechanism	Evidence base	Grade	Observable thesis measure
BDNF/TrkB signaling	Exercise, antidepressant, and psychedelic mechanisms literature links BDNF/TrkB to plasticity	A-D by source	BDNF, behavior, EEG changes
NGF/neuritogenesis	Heridium compounds and extracts show NGF-related neurite effects, with limited human MCI evidence	C/D	Cognitive trajectory and candidate eligibility
Dendritic spine remodeling	Psilocybin and other serotonergic psychedelics show structural plasticity in preclinical models	D	No direct human tract-repair claim
Glymphatic clearance	Sleep state affects metabolite clearance in animal and translational sleep literature	B/D	Sleep efficiency and daytime function
Network reorganization	Stroke/TBI rehabilitation supports surviving-circuit compensation	B	EEG/coherence and functional trajectory

3.5 Psilocybin and neuroplasticity: evidence review

Psilocybin appears in this thesis because it is one of the strongest modern examples of a compound class that can acutely alter large-scale network dynamics and has substantial preclinical support for plasticity-related mechanisms. Human neuroimaging studies demonstrate changes in default-mode and large-scale brain-network organization (Carhart-Harris et al., 2012; Siegel et al., 2024) [Grade B]. Preclinical studies support neuritogenesis, spinogenesis, dendritic spine growth, and TrkB-related plasticity mechanisms (Ly et al., 2018; Shao et al., 2021; Moliner et al., 2023) [Grade D/B mechanism]. Psychiatric trials establish clinical research feasibility and safety-monitoring models but do not establish neurodegenerative efficacy; Goodwin et al. (2022) was a Phase 2b trial, while the COMPASS Phase 3 program results should be treated separately when final citable records are used [Grade A/B]. Controlled-substance governance is specified in Section 5.7 and is not repeated here.

The Lago et al. case report is relevant because it illustrates the latent-capacity hypothesis in advanced dementia while remaining low-level evidence. It is a single case, uncontrolled, and not adequate for causal inference (Lago et al., 2026) [Grade C]. Its role is contextual. It does not prove RAY-VASC; it supports the importance of investigating whether residual function can become transiently accessible under altered network conditions.

Table III-5. Psilocybin evidence table

Study/source	Design	N/population	Intervention/exposure	Measures	Effect direction	Grade	Limitations
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Study/source	Design	N/population	Intervention/exposure	Measures	Effect direction	Grade	Limitations
Carhart-Harris et al. 2012	Human fMRI	Healthy volunteers	Psilocybin exposure in controlled research	DMN/network activity	Altered large-scale network dynamics	B	Acute, not neurodegeneration
Ly et al. 2018	Preclinical/in vitro	Cell and animal models	Classic psychedelics	Neuritogenesis/spinogenesis	Increased structural plasticity markers	D	Translation to humans uncertain
Shao et al. 2021	Mouse in vivo imaging	Mice	Psilocybin	Dendritic spine dynamics	Rapid persistent spine growth	D	Disease relevance unproven
Moliner et al. 2023	Mechanistic/preclinical	Molecular and animal models	Psychedelics and TrkB	BDNF receptor binding/plasticity	Supports the TrkB-mediated plasticity model	D/B mechanism	Clinical disease inference limited
Goodwin et al. 2022	Phase 2b clinical trial	Treatment-resistant depression	Single-dose psilocybin with support	Depression outcomes	Symptom change in the psychiatric population	A/B	Not neurodegeneration; distinct from COMPASS Phase 3 program records
Siegel et al. 2024	Human neuroimaging	Healthy adult research sample	Psilocybin	Brain synchronization/connectivity	Desynchronization and network effects	B	No disease outcomes
Lago et al. 2026	Verified single case report	Advanced Alzheimer's disease	Reported psilocybin-containing mushroom exposure	Functional observations	Transient multidomain improvement reported	C	Verified DOI; no control, no imaging, non-generalizable
Kato et al. 2025	Preclinical aging	Cells and aged mice	Psilocybin exposure	Cell lifespan/survival markers	Longevity signal in model systems	D	Not human clinical evidence

3.6 Neurotrophic mycology and Layer 1 candidate-universe evidence architecture

Section 1.3 defines the candidate universe and participant-specific eligibility filter. This chapter evaluates the evidence classes within that universe; Chapter VI operationalizes eligibility, and Appendix A provides the full compound-level specification.

Table III-6. Candidate-universe analytical summary (abbreviated; full compound-level evidence object in Appendix A).

Candidate class	Representative examples	Evidence center	Grade range	Thesis role	Primary translation barrier/eligibility gate
Functional mycology	Lion's Mane, Cordyceps, Reishi, Turkey Tail, Chaga/Maitake/Shiitake	Human MCI signal for Lion's Mane; stronger preclinical/terrain rationale for several fungi	C-E, with selected D mechanisms	Neurotrophic and terrain hypotheses	Product QA, tolerability, immune/bleeding/renal review, candidate isolation
Cholinergic and memory architecture	Bacopa monnieri, citicoline, Alpha-GPC, phosphatidylserine, huperzine A	Bacopa RCT/meta-analysis support; mixed cognition/stroke literature for selected cholinergic supports	A/B to E	Memory, attention, membrane, and cholinergic-support hypotheses	Cholinergic burden, sleep, heart rate, medication, and slow-onset attribution review
Antioxidant and anti-inflammatory terrain	NAC, alpha-lipoic acid, curcumin, resveratrol, quercetin, omega-3	Human evidence varies by indication; mechanisms are stronger than disease-specific proof	B-E	Oxidative-stress, inflammatory-terrain, and vascular-support hypotheses	Antiplatelet, renal, glucose, GI, and polypharmacy review
Mitochondrial and longevity candidates	CoQ10, PQQ, NR, NMN, Urolithin A	Emerging human and preclinical evidence outside PSP/VCID	B-E	Energy metabolism, mitophagy, and aging-terrain hypotheses	Lab-guided eligibility; renal/metabolic monitoring; no rejuvenation claim

Candidate class	Representative examples	Evidence center	Grade range	Thesis role	Primary translation barrier/eligibility gate
Sleep and circadian supports	Melatonin, magnesium, glycine, L-theanine, apigenin, light timing	Human sleep/circadian evidence varies by candidate and outcome	A-C	Terrain stabilization and confounder reduction	Sedation, falls, next-day cognition, and sleep-clinic context review
Adaptogens and arousal modulators	Ashwagandha, Rhodiola, ginseng, eleuthero, schisandra	Mixed stress/fatigue evidence; limited disease-specific support	B-E	Fatigue, stress, and arousal-terrain hypotheses	BP/HR, agitation, thyroid, sleep, and medication-context review
Future-governed experimental items	Dihexa, 9-MBC, NSI-189, controlled-substance Window items	Preclinical, early-stage, or regulated-context evidence only	D/E unless separately verified	Frontier mapping and exclusion/governance statements	Institutional, legal, IRB, clinician, toxicology, and capacity safeguards only

This structure prevents the thesis from flattening biologically different candidates into a single stack narrative. A memory candidate may operate through cholinergic tone, dendritic complexity, attention, sleep consolidation, inflammation reduction, or vascular stability. A motor or fatigue signal may reflect mitochondrial terrain, respiratory stability, medication timing, sleep quality, rehabilitation intensity, or caregiver prompting. Class differentiation is therefore not cosmetic; it is what makes attribution possible.

Functional mycology is treated as one candidate class, not as a single claim. Lion's Mane is mapped to NGF/neurite and cognitive-support hypotheses; Cordyceps to mitochondrial, fatigue, and respiratory-terrain hypotheses; Reishi and Turkey Tail to immune and inflammatory-terrain hypotheses; and other fungi to conservative polyphenol, metabolic, immune, or quality-control categories. Cholinergic and memory-support candidates form a second class in which *Bacopa monnieri* receives the strongest cognitive-evidence attention, while citicoline, Alpha-GPC, phosphatidylserine, and huperzine A require medication and burden review before any implementation inference.

The Cognitive Turbo concept is handled as a scientific-historical bridge rather than proof. Vedic associations between *Bacopa* and memory remain in a cultural context [Grade E]. The stronger thesis support comes from modern randomized trials and meta-analysis reporting memory or attention signals after sustained *Bacopa* extract exposure in selected populations (Stough et al., 2001; Calabrese et al., 2008; Pase et al., 2012; Kongkeaw et al., 2014) [Grade A/B]. That evidence justifies *Bacopa*'s place in the Layer 1 candidate universe; it does not establish benefit in PSP, VCID, or RAY-001.

The full specification is provided in Appendix A.

3.7 Biomarker rationale: p-tau217, BDNF, NfL, hs-CRP, IL-6, EEG

The biomarker architecture is exploratory but necessary. A thesis about residual adaptive capacity requires objective measures beyond caregiver impression. p-tau217 and NfL support disease-trajectory context. BDNF is an imperfect but relevant peripheral plasticity marker. Inflammatory markers quantify systemic terrain. EEG provides repeated network-state measurement through spectral power, coherence, and event-related potentials such as P3. None of these biomarkers alone proves repair or disease modification. Their value is in convergence across time.

Table III-7. Biomarker evidence and thesis role

Biomarker	Role	Evidence grade	Operational use	Limitation
p-tau217	Tauopathy/AD trajectory context	B	Disease biology and differential pathology	Not PSP-specific standalone proof
BDNF	Plasticity readiness/exploratory marker	B/C	Pre/post terrain marker	Peripheral BDNF variability
NfL	Axonal injury/neurodegeneration burden	B	Exploratory progression marker	Disease non-specific
hs-CRP/IL-6/TNF-alpha	Inflammatory terrain	A/B	Confounder and terrain monitoring	Non-specific systemic signal
EEG spectral power	Network-state/cognitive reserve signal	B/C	Longitudinal repeated marker	Artifact and state dependence
P3 amplitude/latency	Attention/updating marker	A/B	ERP cognitive index	Requires protocol consistency

A full thesis architecture must also distinguish the kinds of absence present in the literature. Some gaps are disease gaps: no PSP-specific psilocybin trial, no VCID-specific Window protocol, no mycology or nootropic trial in PSP-VCID mixed disease, and no biomarker-integrated functional mycology study in such phenotypes. Other gaps are methodological: few protocols combine high-frequency caregiver logging, objective motor/cognitive measures, sleep and respiratory context, medication timing, confounder annotation, and EEG within a single N-of-1 architecture. A third category is translational: many mechanistic candidates remain trapped between preclinical plausibility and human implementation uncertainty. RAY-VASC is designed at the intersection of all three gaps.

The disease gap matters because PSP and VCID have different failure modes. PSP produces progressive tau-associated disruption of ocular motor control, gait, speech, swallowing, executive function, and posture. VCID produces vascular disconnection, attention and executive impairment, gait changes, and stepwise or fluctuating decline. A combined phenotype cannot be adequately studied by borrowing an outcome battery from one disease alone. The outcome system must measure motor severity, executive function, sleep, respiratory status, falls, caregiver burden, and neurophysiology together. A narrow outcome battery would produce cleaner statistics but weaker biological interpretation.

The methodological gap matters because complex phenotypes defeat simple attribution. If a participant improves after a candidate exposure, the improvement could reflect sleep recovery, medication timing, caregiver engagement, regression toward baseline, measurement learning, acute illness resolution, or genuine neurobiological change. If a participant worsens, the worsening could reflect disease progression, a fall, hypoxia, infection, dehydration, medication effects, sleep collapse, or candidate intolerance. RAY-VASC therefore makes confounder capture a core method rather than an appendix.

The translational evidence is uneven. Bacopa has stronger human cognition evidence than many botanicals but remains non-disease-specific; citicoline has mixed and context-dependent findings; Lion's Mane has limited human MCI evidence and a stronger preclinical rationale; and the remaining antioxidant, mitochondrial, sleep, adaptogenic, micronutrient, and fungal candidates occupy distinct evidence and safety lanes.

Bacopa also illustrates why time matters. A compound class that may require sustained exposure before measurable cognitive effects appear cannot be judged by a one-week or four-week window. This temporal feature has direct protocol implications: APPRE must prevent premature abandonment, but it must also prevent indefinite continuation without evidence. The candidate's expected onset, standardization requirements, adverse-effect profile, and interaction context all become part of the response-policy logic. In this sense, the Cognitive Turbo layer is not a promotional label. It is a test case for slow-onset cognitive candidate governance.

The mycology layer requires the same discipline. Lion's Mane is often presented publicly as a cognitive mushroom, but the thesis treats it as a candidate class with limited human evidence and stronger mechanistic/preclinical rationale. Cordyceps is relevant to fatigue, mitochondrial, and respiratory terrain hypotheses, not to proven neurodegenerative repair. Reishi and Turkey Tail are more plausibly immune/inflammatory or gut-immune candidates than direct cognitive interventions. Chaga, Maitake, Shiitake, and related fungi require standardization, contamination, bleeding, immune, and drug-interaction scrutiny before they can be considered in any vulnerable phenotype. The evidence map does not rank candidates by popularity. It ranks them by mechanism, human evidence, safety flags, and interpretability.

Psychedelic neuroplasticity evidence is also not homogeneous. Human psychiatric trials provide safety and efficacy information in selected psychiatric populations but do not prove neurodegenerative benefit. Human neuroimaging studies show changes in network dynamics but do not prove functional recovery. Preclinical studies show cellular, dendritic, or receptor-level plasticity but do not establish human disease modification. Case reports can generate hypotheses about latent capacity but cannot establish causality. The thesis uses these levels separately. It does not stack them into a single strong claim.

The biomarker gap is equally important. A credible residual-capacity thesis cannot rely exclusively on subjective observation. It needs markers that can constrain interpretation, even if no marker is definitive. EEG provides repeated network-state data. p-tau217 can contextualize tau/AD-related biology where relevant, though it is not a PSP-specific standalone marker. NfL can index axonal injury burden, but is not specific. BDNF is relevant to plasticity but noisy in peripheral measurement. Inflammatory markers can show terrain burden but not brain-specific causality. The thesis uses biomarkers as convergence tools, not as proof engines.

3.8 N-of-1 methodology: CENT 2015, design rationale, evidence hierarchy

N-of-1 methodology is appropriate when within-person variability, comorbidity, and confounder density are central to the research problem. CENT 2015 and related reporting standards provide the basis for transparent protocol design, prespecified measures, and interpretable reporting (Vohra et al., 2015) [Grade A]. RAY-VASC uses the N-of-1 framework not as a substitute for trials but as an upstream mechanism-discovery and feasibility architecture. Small-series extension is the next translational step after the single-case architecture is stable.

3.9 Gaps in the literature that RAY-VASC addresses

The literature contains disease-specific descriptions, biomarker studies, psychedelic neuroplasticity work, mycology/nootropic evidence, sleep and exercise evidence, and N-of-1 methodology. It does not yet provide an integrated architecture for PSP-VCID-TBI-sleep-respiratory mixed phenotypes with daily terrain tracking, Layer 1 candidate-universe filtering, APPRE-based interpretability grading, and objective network markers. That gap defines the research space of RAY-VASC.

Chapter IV - Theoretical Framework

The most important gap is not the absence of any single trial. It is the absence of an architecture that can handle heterogeneous mechanisms without collapsing them into a single causal story. PSP trials typically require diagnostic specificity and standardized outcomes. VCID studies often focus on vascular risk, imaging markers, and cognitive trajectories. Psychedelic studies often emphasize psychiatric endpoints or short-term neuroimaging. Mycology and nootropic studies often focus on isolated compounds in healthier or less complex samples. RAY-VASC occupies the space between these literatures: a structured way to determine whether a complex phenotype can generate interpretable, multidomain, longitudinal data before larger trials are justified.

RAY-VASC begins with biological terrain and asks which candidate classes are mechanistically relevant, evidence-supported, eligible for consideration, and measurable enough to interpret. This permits heterogeneous candidates to be compared without treating them as equivalent.

The Cognitive Turbo layer illustrates the distinction between cultural observation, preclinical mechanism, and human trial evidence. *Bacopa monnieri* has a long traditional association with memory and learning, but traditional use is Grade E evidence. Modern *Bacopa* evidence includes randomized trials and meta-analytic reviews showing cognitive effects after sustained use in selected populations (Stough et al., 2001; Calabrese et al., 2008; Pase et al., 2012; Kongkeaw et al., 2014) [Grade A/B]. The thesis treats *Bacopa* as a candidate memory-architecture support with a slow-onset evidence profile, not as a rapid nootropic or disease-specific treatment.

Functional mycology requires similar discipline. Lion's Mane has preclinical and limited human evidence relevant to NGF, neurite outgrowth, and mild cognitive impairment (Mori et al., 2009) [Grade C/D]. That does not establish benefit in PSP or VCID. It establishes a plausible candidate class for evidence mapping and careful safety filtering. Cordyceps, Reishi, Turkey Tail, Chaga, Maitake, Shiitake, and related candidates require even more careful separation between immune, metabolic, respiratory, and neurotrophic claims. A general architecture can map them; a case-specific filter may exclude or defer them.

In this literature context, the thalamus functions as a systems-level bridge between lesion anatomy and observable behavior. Bilateral thalamic infarcts can disrupt arousal, attention, memory relay, motor control, and emotional regulation. The thalamus is not simply a passive relay; it participates in selection, gating, and coordination across cortical and subcortical systems. A thesis about residual capacity must therefore consider not only cortical degeneration but also relay-node disruption. A participant may retain cortical substrate but fail to access it efficiently when thalamic-cortical communication is degraded.

White-matter disease adds a slower disconnection process. Small-vessel injury can reduce processing speed, executive function, gait coordination, and network efficiency without producing a single focal deficit. This is why RAY-VASC emphasizes trajectory and repeated measures. A one-time cognitive score cannot identify whether executive slowing reflects PSP, vascular disconnection, sleep suppression, medication timing, hypoxia, or fatigue. The literature supports the need for integrated interpretation rather than single-diagnosis attribution.

Sleep is the most important reversible terrain candidate in the evidence base. Experimental and translational work links sleep to metabolite clearance, synaptic homeostasis, memory consolidation, arousal, and next-day cognition. In a phenotype with severe PLMD and low sleep efficiency, sleep state can dominate daily outcomes. RAY-VASC therefore treats sleep not as a lifestyle variable but as a primary confounder and potential terrain target.

4.1 The Neuropathic Repair-Remap Hypothesis - four-operation formal model

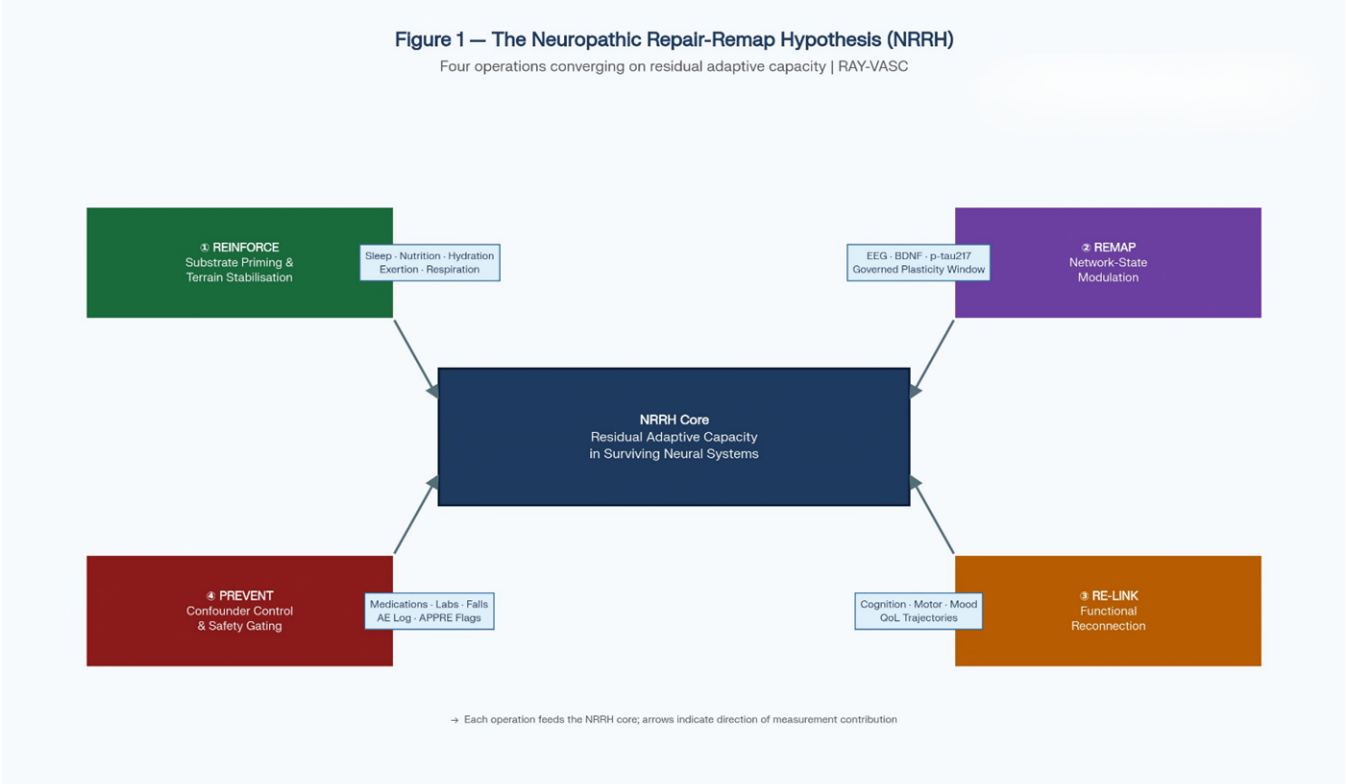


Figure 1. Neuropathic Repair-Remap Hypothesis (NRRH). Four operations — Reinforce, Remap, Re-link, and Prevent — each mapped to their primary measurement domains, converging on the central thesis claim of residual adaptive capacity in surviving neural systems. Arrows indicate the direction of measurement contribution. Author's original framework (Henderson, 2026).

The Neuropathic Repair-Remap Hypothesis [Figure 1] formalizes how surviving neural systems may be studied under structural injury and physiological suppression. It distinguishes permanent tissue loss from potentially modifiable accessibility of viable circuits. The four operations - Reinforce, Remap, Re-link, and Prevent - provide the theoretical bridge from mechanism to measurement.

Reinforcement refers to the stabilization of circuits that remain viable. The relevant mechanisms include BDNF/TrkB signaling, NGF-related neurite support, cholinergic tone, mitochondrial resilience, sleep-driven consolidation, and reduction of inflammatory burden. Remapping refers to compensatory reorganization in surviving networks. Re-linking refers to restored coordination across networks that remain structurally present but functionally under-recruited. Prevention refers to the reduction of destabilizing burdens that accelerate injury or obscure signals.

Table IV-1. NRRH mechanism-to-measurement crosswalk

NRRH operation	Mechanistic thesis	Observable prediction	Primary measures	Falsification signal
Reinforce	Surviving circuits may be stabilized by neurotrophic, cholinergic, metabolic, and sleep-terrain support	Reduced decline rate or improved within-person stability	Daily function, EEG, sleep, and inflammatory markers	No signal despite adherence and clean confounder capture
Remap	Preserved peri-lesional networks may support compensatory structural/functional routing	Task-specific improvement without a global cure claim	Motor/cognitive tests, ERP, gait, caregiver-rated function	No domain-specific change under adequate measurement
Re-link	Large-scale network coordination may become more accessible when the suppression burden falls	Improved coherence/P3 or reduced pathological low-frequency dominance	EEG, P3, cognition, sleep	Network markers worsen or decouple from function

NRRH operation	Mechanistic thesis	Observable prediction	Primary measures	Falsification signal
Prevent	Inflammatory, oxidative, metabolic, sleep, and respiratory destabilizers may accelerate decline and can be monitored	Lower variability and fewer destabilization events	hs-CRP, IL-6, SpO2, sleep, AE log	Terrain markers worsen despite protocol fidelity

4.2 The Regenerative Loop as a multi-domain terrain framework

The Regenerative Loop is the terrain model underlying NRRH. It defines biological domains that are individually important but more important in combination: exertion, sleep/glymphatic function, circadian light, social safety, nutrition/hydration, and respiratory stability. Each pillar can improve or degrade interpretability. A poor night's sleep can suppress cognition and gait. Respiratory instability can mimic neurocognitive decline. Dehydration can alter falls and medication tolerance. Social stress and caregiver burden can change observation quality. The Loop functions as both an intervention terrain and a confounder-control scaffold.

Table IV-2. Regenerative Loop formal model

Pillar	Biological role	Implementation constraint	Measurement variables	Failure mode
Exertion	BDNF, motor learning, vascular health	Falls, fatigue, COPD	PT adherence, TUG, gait speed, dyspnea	Overexertion or injury
Sleep/glymphatic	Clearance, memory consolidation, arousal stability	PLMD/OSA, sedatives	Sleep efficiency, PLMI, awakenings, daytime sleepiness	False cognitive decline
Circadian light	Timing, mood, sleep entrainment	Indoor exposure and mobility	Morning light, bedtime, naps	Rhythm drift
Love/social safety	Stress buffering and adherence support	Caregiver burden	Contact, engagement, affect, caregiver notes	Observer bias or burnout
Nutrition/hydration	Metabolic stability and absorption context	Dysphagia, appetite	Intake, weight, hydration, glucose markers	Malnutrition/dehydration
Respiratory stability	Oxygenation and exertion tolerance	COPD and nebulizer use	SpO2, dyspnea, rescue therapy	Hypoxic confounding

4.3 Substrate priming: the precondition for Governed Plasticity Window science

Substrate priming is the thesis term for preparing biological terrain before interpreting plasticity-sensitive changes. It does not mean that supplements, mushrooms, sleep, exertion, or social routines are proven disease-modifying therapies. It means that unstable terrain makes the Window research uninterpretable. A participant with severe sleep fragmentation, medication variability, respiratory distress, and fluctuating intake cannot provide a clean signal unless those variables are measured and stabilized as much as feasible.

Candidate classes function as a graded hypothesis space; participant-specific eligibility is operationalized in Chapter VI and Appendix B.

4.4 The Three-Phase Architecture: Pre-Load -> Governed Plasticity Window -> Sustain and Integration

The Three-Phase Architecture [Figure 2] converts the theoretical model into a temporal research design. The Pre-Load phase establishes baseline, safety, adherence, terrain stabilization, and candidate eligibility. The Window phase is conditional and external to routine use; it refers only to lawful, clinician-governed, ethics-approved research contexts in which Window exposures could be studied. The Sustain and Integration phase tracks durability, decay, and confounder interaction over time.

Table IV-3. Three-Phase architecture and thesis prediction

Phase	Purpose	What it can test	What it cannot claim	Gate
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Phase	Purpose	What it can test	What it cannot claim	Gate
Pre-Load	Stabilize terrain and build a measurement baseline	Feasibility, adherence, sleep/metabolic change, and candidate tolerability	Disease modification	Clean baseline and safety review
Governed Plasticity Window	Study whether a lawful clinician-governed plasticity exposure changes the network trajectory	Signal timing, safety, EEG/functional changes	Self-directed use, efficacy, repair	External ethics/legal/clinical oversight
Sustain and Integration	Track durability and prevent attribution collapse	Persistence, decay, confounder interaction	Cure or permanent reversal	Longitudinal adherence and repeated measures

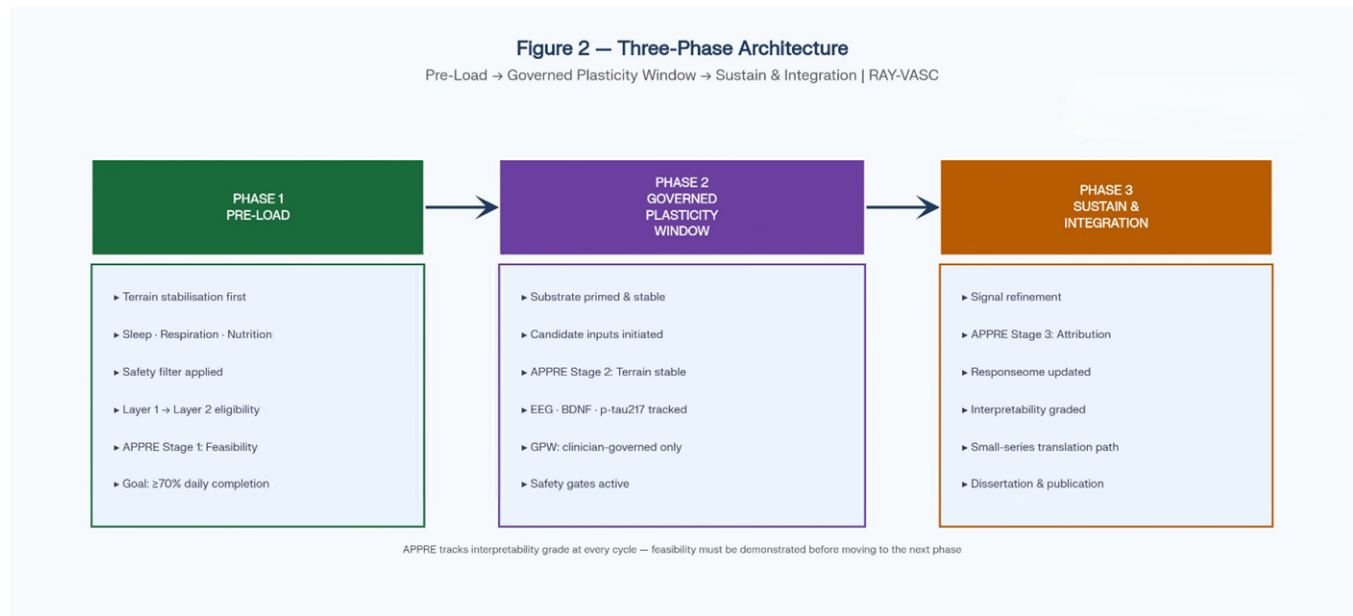


Figure 2. Three-Phase Architecture: Pre-Load, Governed Plasticity Window, and Sustain & Integration. Phase objectives and APPRE stage alignment are shown for each phase. No phase transition is permitted without satisfying the prior phase's feasibility threshold. GPW = Governed Plasticity Window; APPRE = Algorithmic Personalisation Predictive Response Engine. Author's original framework (Henderson, 2026)

The NRRH also requires a theory of sequence. A damaged system should not be challenged with complex exposures before the measurement environment is stable. The Pre-Load phase is therefore not a preparatory ritual; it is a validity condition. If sleep is catastrophically fragmented, if falls are unlogged, if medication timing is unstable, if respiratory status is unknown, and if caregiver reporting is inconsistent, then a later change cannot be interpreted. The first research task is to reduce noise and document residual variability.

The substrate-priming concept is sometimes misunderstood as a claim that the stack repairs the brain before a Governed Plasticity Window. The thesis makes a narrower statement. Substrate priming aims to improve the conditions under which surviving circuits can be measured. It may support neurotrophic signaling, membrane integrity, cholinergic function, sleep consolidation, metabolic stability, inflammatory reduction, or rehabilitation readiness, depending on the candidate and the evidence grade. The observable question is whether those conditions improve in measurable ways and whether improved terrain corresponds to more stable function.

The governed Window phase has a different role. Its scientific premise is that some neuromodulatory exposures can acutely alter network dynamics and may open a time-limited period of enhanced plasticity. In RAY-VASC, this remains conditional future research. It is not necessary for Phase 1 feasibility, not part of a home protocol, and not interpreted without independent governance. The architecture can succeed at the feasibility and terrain levels even if no Window substudy is ever performed.

The Sustain and Integration phase is necessary because transient changes are not enough. A signal that appears briefly and disappears without explanation is scientifically weak unless it is captured with timing, context, and objective measures. Integration asks whether any observed change stabilizes, decays, or depends on continued terrain support. It

also asks whether gains in one domain transfer to another domain or remain task-specific. This is how the architecture avoids confusing acute state change with durable adaptation.

The theory also places boundaries around the term network. Network language can become vague if not tied to measures. In this thesis, network hypotheses correspond to EEG spectral organization, P3 amplitude/latency, functional performance, sleep state, and domain-specific behavior. Future imaging could strengthen network inference, but the current architecture does not require fMRI. The research question is whether feasible objective markers can be collected repeatedly enough to constrain interpretation.

Falsification is built into each theoretical layer. If Pre-Load improves logs but not terrain markers, substrate-priming claims weaken. If terrain improves but function does not stabilize, the residual-capacity interpretation weakens. If the function changes without objective marker correspondence, the network interpretation weakens. If all apparent changes occur during confounded cycles, APPRE rejects them as weak evidence. This structure prevents the theory from being protected by narrative flexibility.

4.5 The dam-release model: scientific translation and failure conditions

The dam-release model is a metaphor for a testable sequence: substrate priming may reduce suppressive load, and a governed Window exposure may produce measurable changes in network coordination if surviving circuits retain adaptive capacity. The metaphor is useful only if operationalized. In RAY-VASC, operationalization means baseline measurement, sleep and respiratory context, candidate eligibility, EEG, functional outcomes, biomarkers, adverse-event capture, and APPRE interpretability grading. If those measures do not show convergent change, the metaphor is weakened.

Terminology follows the definitions in the List of Abbreviations and Key Terms.

4.6 Convergence: how NRRH, the Loop, and Three-Phase logic integrate

The theoretical system converges through a sequence. NRRH defines what may be measurable. The Regenerative Loop defines the terrain that must be stabilized and tracked. The Three-Phase Architecture defines when different research questions can be asked. APPRE defines how observed changes are interpreted without collapsing mechanism, safety, and outcome into one undisciplined narrative. The framework is structured to permit falsification, evidence grading, and protocol-level audit by external reviewers.

Chapter V - Computational Framework: APPRE

Chapter V formalizes APPRE as the computational-methods layer of the thesis. The preceding chapters establish the disease context, evidence base, and theoretical claim. APPRE defines how those claims are made observable without allowing candidate enthusiasm to outrun safety, data quality, or public evidence grade. Its function is not to automate clinical judgment. Its function is to prevent unstructured interpretation from masquerading as scientific inference.

APPRE stands for Algorithmic Personalization Predictive Response Engine. In this thesis, the term algorithmic means rule-bound, auditable, and preregisterable. It does not mean black-box machine learning, validated medical-device software, or clinical decision support. APPRE is a schema-and-rules framework for deciding whether a cycle is eligible for interpretation, whether a candidate is eligible for observation, whether a safety event requires a hold, and whether a policy update is documented before or after outcome review. Its central scientific contribution is interpretability governance.

5.1 Why APPRE exists: the interpretation failure problem in N-of-1 research

Complex N-of-1 research fails when a single observed change is treated as an intervention effect. In RAY-001, cognition may change because sleep improves, medication timing shifts, respiratory burden changes, a fall occurs, caregiver prompting increases, pain changes, hydration improves, infection resolves, or measurement familiarity develops. A motor score may change because of levodopa timing, fatigue, orthostasis, fear of falling, footwear, respiratory status, or day-to-day sleep fragmentation. APPRE exists because none of these confounders is peripheral. They are the terrain in which the outcome occurs.

APPRE therefore moves the research object from intervention exposure to an interpretable cycle. A candidate exposure is only meaningful when it is embedded in a cycle with recorded adherence, safety status, medication timing, sleep context, respiratory context, adverse-event status, and outcome timing. A clean non-exposure cycle can be more informative than a poorly logged exposure cycle. This is the central logic that distinguishes RAY-VASC from a supplement diary.

Table V-1. APPRE rule-set.

Domain	Input fields	Core rule	Output policy	Practical novelty
Subject profile	Diagnosis, age, medications, comorbidities, caregiver capacity, swallowing status, fall risk	No candidate is considered without phenotype and capacity fit	Eligible, hold, excluded, or clinician-review status	Requires phenotype and capacity fit before exposure selection
Safety	Renal trend, antiplatelet context, BP/HR, COPD status, falls, dysphagia, sleep instability	Safety veto overrides mechanistic interest	Hold, exclude, simplify, or escalate	Imposes a formal safety veto over mechanistic interest
Exposure	Candidate, class, start/hold/continue/stop status, adherence, formulation, delivery method	Unlogged or non-isolated exposure is not interpretable	Downgrade the cycle grade or repeat after isolation	Treats formulation and delivery as part of actual exposure
Outcome	Cognition, motor, sleep, mood, respiratory, caregiver, EEG, biomarkers	Outcome change requires a matched confounder context	Trajectory classification only after context review	Links every outcome to the matched context before classification
Confounder	Sleep disruption, medication change, fall, illness, hydration, caregiver burden, acute stress	Confounders annotate every candidate and outcome window	Clean, qualified, or uninterpretable cycle	Makes cycle quality an explicit analytical variable
Policy update	Weekly/monthly review, versioned rule set, amendment log	Policy cannot change retrospectively after seeing the desired outcomes	Continue, pause, simplify, hold, exclude, or clinician-review	Creates prospective, versioned, non-retroactive adaptation logic

Figure 3 — APPRE: Inputs, Filters, Responseome & Policy Decision
Algorithmic Personalisation Predictive Response Engine | RAY-VASC

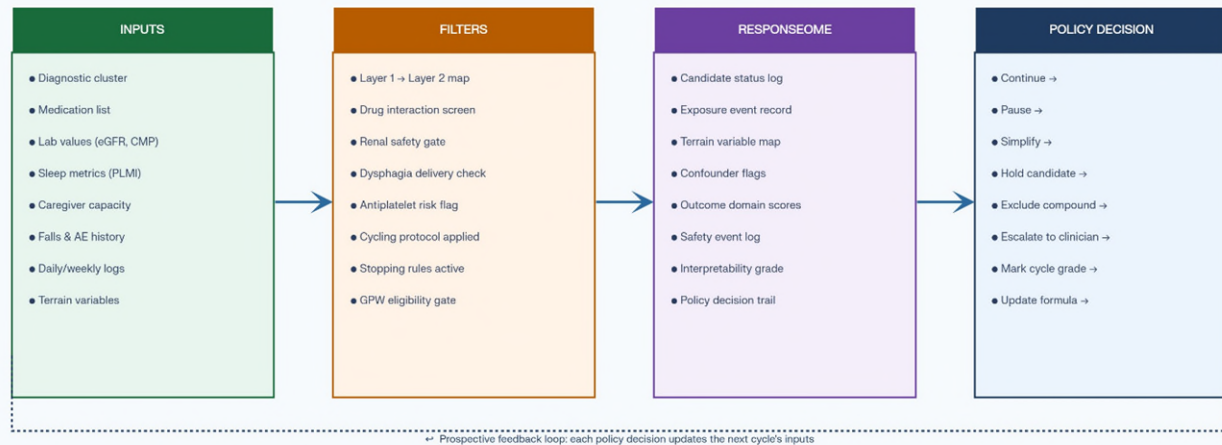


Figure 3. APPRE (Algorithmic Personalisation Predictive Response Engine): four-stage operational flow. Inputs (diagnostics, medication, labs, daily logs) pass through safety Filters (drug interactions, renal gate, stopping rules), populate the Responseome (confounder-annotated outcome and exposure record), and generate a prospective Policy Decision. A feedback loop updates the next cycle's inputs. Author's original schema (Henderson, 2026).

5.2 The APPRE loop: profile -> eligibility -> exposure -> outcome -> confounder -> safety -> interpretability -> policy update

APPRE [Figure 3] begins with the subject profile and ends with a documented policy statement. The loop is deliberately slow. Profile defines disease and risk context. Eligibility separates Layer 1 evidence candidates from Layer 2 participant-specific implementation. Exposure logging records whether a candidate was actually present in a defined cycle. Outcome measurement records domain movement. Confounder capture determines whether movement can be interpreted. Safety review determines whether continuation is permissible. Interpretability grading classifies the cycle. Policy update records the next rule-bound step.

The loop is recursive but not self-authorizing. A favorable outcome cannot authorize a candidate who violates safety rules. A public thesis cannot convert a source-pending lead into a clinical claim. A cleaner trajectory cannot override clinician oversight. In this way, APPRE is not merely a data structure; it is the governance logic that keeps a high-ambition thesis from becoming an uncontrolled intervention narrative.

5.3 Interpretability grading: the core innovation

APPRE grades cycles as clean, qualified, or uninterpretable. A clean cycle has adequate adherence, stable medication timing, documented sleep and respiratory context, no major acute illness, no major safety event, and available outcome data. A qualified cycle contains meaningful confounders but remains analytically usable if those confounders are explicit. An uninterpretable cycle has missing exposure data, insufficient outcome data, simultaneous changes, safety instability, or uncontrolled confounders.

Interpretability grading also protects negative findings. If a candidate produces no change during clean cycles, that is useful evidence. If a candidate appears favorable only during confounded cycles, that is weak evidence. If a safety hold interrupts candidate exposure, the hold is not a protocol failure; it is a protocol function. APPRE makes this visible by treating safety interruptions, missingness, and null signals as first-class research outputs.

5.4 Three-stage optimization architecture

APPRE optimization is not a search for maximal input intensity. It is a staged process that prioritizes feasibility, terrain stabilization, and signal refinement. The system cannot move to signal refinement until feasibility and safety are

established. It cannot interpret candidate effects until terrain and confounders are documented. It cannot support small-series translation until the rules set can be used by reviewers other than the originator.

Table V-2. Three-stage optimization logic.

Stage	Purpose	Rule	Minimum threshold	Failure signal
Stage 1 - Feasibility	Prove that the case can generate reliable data	No complexity added until core logging is stable	$\geq 70\%$ daily/weekly completion target	Missingness, caregiver burden, unsafe testing
Stage 2 - Terrain stabilization	Reduce reversible suppression and dominant confounding	Prioritize sleep, respiratory, medication-timing, hydration, exertion, and safety context	Consecutive clean or qualified cycles	Persistent instability prevents attribution
Stage 3 - Signal refinement	Compare within-person trajectories under rule-bound exposure states	Candidate changes require isolation, monitoring, and predeclared outcome windows	Repeated directional signal across ≥ 2 interpretable cycles and ≥ 2 domains when feasible	Single-domain anecdote, simultaneous changes, or safety interruption

5.5 The formula-intelligence goal and the responseome

APPRE's long-term contribution is formula intelligence: the ability to document which candidate classes, terrain variables, safety filters, and outcome domains cluster together across related phenotypes without pretending that early N-of-1 data can produce universal treatment rules. The unit of learning is the responseome, defined as the participant-specific longitudinal map of exposure status, terrain variables, confounders, safety events, outcome trajectories, interpretability grades, and policy decisions.

A responseome is not a clinical identity and not a validated prediction engine. It is a structured research object. In RAY-VASC, it bridges one participant's time series and future small-series comparison. The same participant may have sleep-dominant, medication-dominant, respiratory-dominant, or candidate-associated cycles. APPRE's role is to preserve those distinctions so the thesis does not collapse everything into one causal story.

Table V-3. Responseome schema.

Field	Type	APPRE source	Analytical use	Small-series extension path
Participant identifier	De-identified string	Subject profile	Links observations without exposing identity	Enables cross-case grouping without raw identifiers
Diagnostic cluster	Categorical	Subject profile	Stratifies PSP, VCID, TBI, sleep, respiratory, vascular, and mixed phenotypes	Defines subgroup architecture
Candidate status	Eligible, held, excluded, clinician-review, active, stopped	Eligibility and safety rules	Prevents Layer 1 candidates from being confused with active exposures	Allows comparison of exclusion logic across cases
Exposure event	Time-stamped structured field	Daily/weekly log	Anchors candidate timing to outcomes and confounders	Supports cycle-level comparison
Formulation and delivery	Capsule, powder, liquid/suspension, crushed-capsule vehicle, food-associated, not applicable	Exposure log and dysphagia field	Document the adherence feasibility and absorption/tolerability context	Standardizes dysphagia-sensitive documentation
Terrain variables	Numeric/categorical	Sleep, exertion, respiration, hydration, nutrition, light, social context	Separates the biological state from the candidate effect	Identifies terrain-dominant response patterns
Confounder flags	Binary/severity-coded	Confounder matrix	Downgrades or qualifies the interpretation	Quantifies clean-cycle proportion
Outcome domain scores	Numeric/instrument-specific	Outcome battery	Tracks cognitive, motor, mood, sleep, respiratory, caregiver, EEG, and biomarker trajectories	Enables shared domain-level modeling
Safety events	Severity-coded	AE/stopping-rule log	Determines holds, exclusions, and escalation	Maps phenotype-risk relationships

Field	Type	APPRE source	Analytical use	Small-series extension path
Interpretability grade	Clean, qualified, uninterpretable	APPRE review	Defines whether a cycle can support inference	Harmonizes cycle quality across cases
Policy decision	Continue, pause, simplify, hold, exclude, escalate	APPRE policy update	Records transparent response-policy logic	Enables audit of adaptive decisions

5.6 APPRE as a methodological contribution

APPRE contributes a formal link between individualized exposure tracking and public research audit. It requires the investigator to state what was eligible, what was excluded, what changed, what confounded the cycle, what safety events occurred, and what policy decision followed. This structure reduces post-hoc narrative drift and makes external replication possible.

The formula-intelligence layer is especially important for future replication. If several participants share mixed neurovascular phenotypes but differ in medications, renal status, sleep burden, caregiver capacity, respiratory disease, dysphagia, and falls, identical exposure rules would be scientifically crude. APPRE permits the rule logic to remain comparable while the eligible candidates differ. The shared object is not on the same stack. The shared object is the same process: profile, filter, expose, measure, annotate, grade, and update.

5.7 Controlled-substance boundary

The controlled-substance boundary belongs in APPRE because APPRE is the point where eligibility, exposure records, policy logic, safety rules, and governance constraints meet. The governed Window concept is a research-design construct, not a home-use protocol, and not a dosing or sourcing guide. APPRE cannot recommend, tune, source, prepare, administer, or direct controlled-substance exposure. It also cannot convert observational data into permission for self-directed use.

Any future Windows sub-study would require independent legal access, ethics approval, clinician governance, capacity assessment, psychiatric and cardiovascular screening, medication review, adverse-event planning, and documented consent safeguards. In RAY-001, cognitive impairment, falls, COPD, sleep instability, polypharmacy, dysphagia, and caregiver burden would all require conservative exclusion or institutional review before any such study could be considered. The thesis treats psilocybin as a literature and future-substudy object only. It is included because of its relevance to network-state science and neuroplasticity mechanisms, not because it is an active intervention layer in the public protocol.

Table V-4. APPRE failure mode registry (main-text summary; full specification in Appendix D).

Failure class	Core risk	APPRE response	Full response specification
Attribution error	A confounder is mistaken for a candidate effect	Downgrade interpretability; require repeat or stabilization	Appendix D and Appendix E
Safety overreach	Mechanistic interest overrides risk profile	Hold, exclude, simplify, or escalate	Appendix B, D, and F
Protocol creep	Multiple changes collapse causal interpretability	Mark cycle qualified/uninterpretable; simplify next cycle	Appendix D and H
Caregiver-burden failure	Data capture exceeds household capacity	Reduce to Tier 1 logging and record feasibility failure	Appendix C and H
Delivery ambiguity	Dysphagia changes actual exposure	Record delivery method, vehicle, texture, percent consumed, and adverse event	Appendix H

Table V-5. APPRE controlled-substance boundary (summary; full specification in Appendix D).

Boundary	Rule	Allowed thesis treatment
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Boundary	Rule	Allowed thesis treatment
Controlled substances	APPRE cannot tune, recommend, source, prepare, administer, or instruct use	Published evidence, governance requirements, and future-study conditions only
Clinical decisions	APPRE cannot override licensed clinicians	Medication timing logs, adverse-event documentation, and clinician-review flags
Capacity and consent	Any future substudy requires independent safeguards	Eligibility criteria and consent documentation only
Safety events	Safety interrupts interpretation	AE log, hold status, and escalation path
Public release	The thesis is an evidence and methods document	No treatment manual, dosing guide, or home-use protocol

Table V-6. Anti-tolerance and delivery-method cycling protocol.

Candidate class	Why cycling/delivery tracking matters	Recorded field	Safety interpretation
Cholinergic candidates	Cumulative ACh burden and attribution collapse	Start/hold/continue/stop; delivery form	Medication review required
Adaptogens	Sedation, stimulation, BP, sleep, and tolerance effects	On/off window, timing, BP/HR flags	Hold if arousal or cardiovascular state destabilizes
Sleep agents	Sleep improvement can be confused with daytime sedation	Timing, next-day sedation, falls	Sleep signal and safety signal are co-recorded
Polyphenol/antioxidant candidates	Interaction, GI, renal, and antiplatelet context	Food/fat co-administration, labs, bleeding flags	Safety-gated terrain candidate
Functional mycology	Standardization and tolerability affect interpretation	Lot/source QA, powder/capsule/food vehicle	Immune/allergy/bleeding flags
RAY-001 dysphagia variable	Modified delivery changes exposure validity	Texture, vehicle, percent consumed, cough/choke event, caregiver assistance	Delivery is treated as adherence, safety, and confounder data

Full specifications are provided in Appendices D, E, F, and H.

Chapter VI - Methodology

Chapter VI defines how the thesis becomes an auditable research program. The methodology is intentionally more conservative than the private mechanistic intuition. RAY-VASC does not begin by asking whether a stack works. It begins by asking whether a complex phenotype can be characterized, whether eligible candidate inputs can be separated from excluded candidates, whether safety can be documented, whether outcomes can be measured with enough frequency, and whether data can be graded for interpretability before any signal claim is made.

6.1 Study design: N-of-1 longitudinal observational architecture

The RAY-VASC design is prospective, longitudinal, observational, and initially N-of-1 with a small-series translation path. The epistemic target is not randomized efficacy. The target is feasibility, safety documentation, trajectory resolution, confounder control, and objective-signal development in a complex phenotype. CENT 2015 and related reporting standards provide the reporting logic for transparent N-of-1 methods (Vohra et al., 2015) [Grade A].

The primary study objects are not cures, treatment effects, or disease-modifying claims. They are completion rate, adverse-event capture, baseline stability, confounder density, clean-cycle proportion, trajectory direction, signal durability, and replication readiness. A candidate can remain biologically interesting while being ineligible for a participant. A participant can generate important feasibility data without demonstrating functional improvement. These distinctions are central to the methodology.

6.2 Participant eligibility and case characterization reference

Eligibility is determined by diagnostic characterization, caregiver capacity, safety profile, and measurement feasibility. The originating case is described in Chapter II because case complexity determines protocol burden. Future small-series participants would require related mixed neurovascular-neurodegenerative phenotypes, capacity and proxy-consent safeguards where applicable, caregiver feasibility, and clinician oversight. The public thesis does not specify a treatment pathway. It specifies the research architecture required before any intervention pathway can be responsibly evaluated.

6.3 Intervention description and eligibility architecture

The methodology applies the two-layer distinction defined in Section 1.3: the evidence map identifies candidate classes, and participant-specific filtering determines eligibility. Table VI-1 operationalizes that transition.

Table VI-1. Layer 1 -> Layer 2 protocol eligibility crosswalk.

Candidate/class	Layer 1 evidence role	RAY-001 Layer 2 status	Primary exclusion/hold rationale	Monitoring required if conditional	Protocol interpretation
Lion's Mane	Functional mycology; NGF/neurite plausibility; small human MCI evidence	Conditional	Product standardization, allergy/GI tolerance, attribution	Tolerability, GI symptoms, lot/source QA, cognitive schedule	Candidate of interest, not efficacy claim
Cordyceps	Respiratory/mitochondrial fatigue rationale	Conditional/clinician-review	COPD, heart-rate/stimulation attribution, exertion confounding	SpO2, dyspnea, rescue-use log, HR, fatigue	Terrain candidate; must be separated from respiratory variability
Reishi	Inflammation/immune modulation rationale	Hold/clinician-review	Bleeding/immune flags with antiplatelet/falls context	Bruising/bleeding, immune/infection status, clinician review	Not active without safety clearance
Turkey Tail	Beta-glucan/gut-immune candidate	Conditional	GI/immune attribution; weak direct neuroplasticity evidence	GI tolerance, infection context, adherence	Terrain/supportive candidate only

Candidate/class	Layer 1 evidence role	RAY-001 Layer 2 status	Primary exclusion/hold rationale	Monitoring required if conditional	Protocol interpretation
Chaga/Maitake/Shiitake	Immune/metabolic/polyphenol rationale	Hold unless QA clears	Contamination, oxalate/renal, immune attribution	Product QA, CMP/eGFR, GI status	Evidence-map object; not active default
Bacopa monnieri	Cognitive Turbo; memory/attention evidence base	Conditional	Slow onset, GI/sedation, attribution window	Standardization, adherence, sedation, GI effects, cognitive schedule	Requires 8-24 week interpretation horizon
Citicoline	Cholinergic/membrane candidate	Conditional	Activation/headache/sleep attribution; medication context	Sleep, headache, agitation, medication timing	Candidate to isolate slowly
Alpha-GPC	Cholinergic precursor	Hold/clinician-review	Cholinergic burden and attribution risk	Cognition, sleep, HR/BP, clinician review	Not active without review
Huperzine A	AChE inhibition	Excluded/pending	Long half-life, cholinergic burden, and medication interaction concern	Not active	Formal clinician review only
NAC	Antioxidant/glutathione support	Conditional/clinician-review	Bleeding, GI, renal variability	Bruising, GI symptoms, CMP/eGFR	Safety-gated oxidative-stress candidate
Alpha-lipoic acid	Mitochondrial/redox/metabolic support	Conditional	Glucose/renal/GI attribution	CMP/eGFR, glucose symptoms, GI tolerance	Conditional terrain/metabolic candidate
Curcumin/Omega-3 high-dose	Anti-inflammatory/vascular rationale	Hold/clinician-review	Bleeding concern with antiplatelet/falls context	Bruising/bleeding log, clinician review	Not active unless safety-cleared
Sleep/circadian supports	Terrain stabilization	Conditional	Sedation, morning falls, cognition	Sleep diary, morning sedation, falls, caregiver report	Terrain-first can confound all domains
Adaptogens/stimulatory botanicals	Fatigue/stress terrain	Hold/clinician-review	BP, sleep, agitation, stimulation, thyroid/medication attribution	BP/HR, sleep, agitation, falls	Not an active default in a high-burden case
NR/NMN/Urolithin A	NAD/mitochondrial/aging terrain	Pending	Renal/metabolic monitoring and evidence limits	CMP/eGFR, A1c, tolerability	Future candidate only
Experimental neurotrophics	Research-only mechanistic leads	Excluded	Regulatory and safety constraints	Not active	IND/IRB/institutional governance only
Psilocybin	Network-state/plasticity literature object	Not active; future governed substudy only	Legal, psychiatric, cardiovascular, capacity, falls, COPD, dysphagia, polypharmacy	Institutional screening has only ever been studied	Literature/future-study object, not APPRE-tuned protocol
Rehabilitation and Regenerative Loop variables	Terrain and integration layer	Active observational capture	Burden and safety constraints	PT/speech logs, exertion, light, sleep, hydration, social contact	Core measurement variables rather than supplement candidates

6.4 Outcome measures

The outcome system includes primary, secondary, and exploratory measures. Primary measures prioritize feasibility, safety, adherence, sleep, motor function, and cognitive trajectory. Secondary measures include mood, caregiver observation, respiratory status, and rehabilitation variables. Exploratory measures include EEG and biomarkers. Each measure is interpreted in context, not alone.

Table VI-2. Full outcome measure battery.

Domain	Construct	Instrument/measure	Frequency	Tier	Primary use
Feasibility	Completion and burden	Daily/weekly completion, missingness map, caregiver burden note	Daily/weekly	Tier 1	Primary feasibility endpoint
Safety	Adverse events and holds	AE log, falls/head impact, bleeding, respiratory, renal, behavior, dysphagia events	Continuous/event-driven	Tier 1	Primary safety endpoint
Cognition	Global cognition and executive function	MoCA, Trail Making A/B, verbal fluency	Baseline/monthly or feasible interval	Tier 1/2	Functional trajectory
Motor/gait	Mobility and Parkinsonism	Timed Up and Go, 10-meter walk, MDS-UPDRS III or PSPRS subset	Weekly/monthly	Tier 1/2	Motor trajectory and fall-risk context
Sleep	Sleep architecture and fragmentation	Sleep diary, PSQI, wearable metrics, CPAP/PLMD treatment status	Daily/weekly/monthly	Tier 1	Dominant confounder and intervention terrain
Respiratory	COPD terrain and oxygenation	SpO2, mMRC/CAT, dyspnea notes, rescue/nebulizer use	Daily/weekly	Tier 1	Safety and fatigue context
Mood/behavior	Depression, anxiety, apathy, agitation	PHQ-9, GAD-7, caregiver behavior notes	Weekly/monthly	Tier 2	Secondary functional context
Speech/swallow	Dysphagia and communication	Diet texture, cough/choke events, speech therapy notes, delivery-method exposure field	Daily/event-driven	Tier 1	Safety, adherence, and exposure validity
EEG	Network state	Resting spectral power, coherence, P3 amplitude/latency, artifact rate	Scheduled Tier 2 sessions	Tier 2	Objective network marker
Biomarkers	Inflammation, plasticity, disease context, metabolic/renal status	BDNF, p-tau217, NfL, hs-CRP, IL-6, CMP, A1c, B12/homocysteine	Baseline and intervals	Tier 2/3	Exploratory mechanistic context
Caregiver observation	Daily function and burden	Structured caregiver notes and feasibility ratings	Daily/weekly	Tier 1	Proxy function and implementation realism

6.5 Safety monitoring protocol

Safety monitoring precedes signal interpretation. Falls, head injury, respiratory instability, bleeding, renal change, severe sleep disruption, behavioral destabilization, medication changes, dysphagia events, and caregiver burden trigger review. Nonessential research changes can be paused without treating the pause as a protocol failure.

Table VI-3. Safety filter matrix.

Safety domain	RAY-001 risk factor	Affected candidate/measure, classes	Required monitoring	Action path
Renal variability	eGFR around 65 with documented nadir 44	NAC, ALA, magnesium, mushroom concentrates, NAD-pathway candidates, polypharmacy load	CMP/eGFR, hydration, acute illness status	Hold renal-relevant candidates during instability; clinician review
Antiplatelet/falls context	Clopidogrel history plus fall risk	Curcumin, high-dose omega-3, reishi, anticoagulant-interacting candidates	Bruising, bleeding, falls/head impact	Exclude or hold flagged candidates; clinician review
COPD/respiratory burden	COPD, rescue/nebulizer context, fatigue	Cordyceps, sedating sleep agents, exertion variables, and motor tests	SpO2, dyspnea, rescue use, infection symptoms	Pause testing during flare; mark cycle qualified/uninterpretable
Dysphagia	Soft/pureed diet and swallowing constraints	Capsules, powders, tablets, liquid/suspension exposure records	Texture, vehicle, percent consumed, cough/choke event, caregiver assistance	Treat delivery method as exposure variable; hold unsafe forms
Falls and head injury	Recurrent falls and trauma history	Motor tests, sedating agents, BP-affecting candidates, sleep agents	Fall log, head impact, orthostasis, assistive-device status	Stop testing after high-risk events; clinical evaluation
Medication timing	Sinemet, trazodone, BP medications, and respiratory medications	Motor/cognition/sleep outcomes and candidate attribution	Daily medication timing and missed/changed doses	Anchor tests to medication state; downgrade cycles if unstable

Safety domain	RAY-001 risk factor	Affected candidate/measure, classes	Required monitoring	Action path
Cognitive capacity	MoCA 11/30 baseline	Consent, self-report reliability, complex testing	Proxy support, clinician guidance, comprehension status	Use proxy/caregiver logs; avoid unsupervised high-risk decisions
Caregiver burden	Proxy-dependent data capture	All high-frequency forms and safety logs	Completion rate, late entries, burden rating	Simplify to Tier 1 if burden rises

6.6 Confounder capture and analytical control

Confounder capture is not optional. Sleep, medication, falls, respiratory status, illness, hydration, caregiver availability, diet, testing environment, and acute stress can all distort outcome interpretation. APPRE preserves these variables as analytic metadata rather than diary clutter.

Table VI-4. Confounder control matrix.

Confounder	Signal distortion risk	Capture method	Exclusion or flag window	Statistical/interpretive handling
Sleep disruption	Suppresses cognition, gait, mood, EEG, and caregiver-rated function	Sleep diary, wearable, PSQI, CPAP/PLMD treatment status	Same day and next day flagged; severe disruption can invalidate testing	Sensitivity analysis excluding poor-sleep days; annotate all outcomes
Medication timing change	Alter motor, cognition, alertness, orthostasis, sleep	Daily medication timing log and change notes	Affected cycle flagged	Stratify by medication state; do not compare off-state with on-state as equivalent
Falls/head injury	Acute neurologic and functional effects	Event log, head impact field, clinician review	Until clinically stable	Exclude from candidate inference or treat as a safety endpoint
Respiratory exacerbation	Hypoxia/fatigue effect on all functional measures	SpO2, dyspnea, rescue-use/nebulizer log, infection symptoms	Acute illness/flare window	Exclude or qualify cycle; analyze respiratory context separately
Infection/systemic inflammation	Sickness behavior, fatigue, biomarker distortion	Symptoms, temperature, WBC/CRP if available	Acute illness and recovery window	Annotate/invalidate cycle; avoid biomarker overinterpretation
Hydration/nutrition	Performance, orthostasis, absorption, GI tolerance	Daily intake notes and meal/timing field	Same-day flag	Context annotation and exposure-validity note
Dysphagia/delivery changes	Alters actual exposure and safety	Delivery-method field, texture, vehicle, percent consumed, cough/choke	Event day and the next day if symptomatic	Treat as an exposure modifier, not a minor note
Caregiver availability	Missingness and observer bias	Completion timestamp, burden rating, proxy identity if applicable	High-burden periods flagged	Missingness heat map; simplify packet if persistent
Rehabilitation/exertion changes	Motor/cognition changes are independent of candidate exposure	PT/speech/exertion log	Same day and next day flagged	Annotate trajectory; separate rehab-dominant cycles
Testing environment/practice	Familiarity, fatigue, distractions, time-of-day effects	Test condition note, time, assistance level	Affected test flagged	Maintain consistent conditions; interpret trend, not single value

Table VI-5. Stopping rules and safety escalation.

Trigger	Threshold	Immediate action	Responsible party	Research data consequence
Fall with head impact, loss of consciousness, new neurologic deficit, or unexplained acute decline	Any event	Stop research changes; seek clinical evaluation	Caregiver/clinician/emergency services as appropriate	Cycle marked safety event; candidate inference suspended
Acute respiratory distress, sustained concerning oxygenation, or COPD flare	Clinically concerning symptoms or SpO2 trend	Hold nonessential protocol tasks; follow the medical plan	Caregiver/clinician	Respiratory window excluded or qualified

Trigger	Threshold	Immediate action	Responsible party	Research data consequence
Bleeding/bruising concern	New unexplained bruising, bleeding, black stool, or clinician concern	Hold bleeding-flagged candidates; clinician review	Caregiver/clinician	Safety endpoint; exposure cycle qualified/uninterpretable
Renal decline or abnormal CMP	Clinically meaningful eGFR drop, dehydration, AKI concern, abnormal creatinine/BUN	Hold renal-relevant candidates and unnecessary supplement complexity	Clinician	Candidate expansion is prohibited until stable
Severe sleep worsening or daytime somnolence	Sustained insomnia, hypersomnolence, confusion, or morning fall risk	Pause new sleep agents/inputs; review sleep plan	Caregiver/clinician	Sleep confounder flag; outcome interpretation downgraded
Behavioral destabilization	Hallucination, agitation, unsafe impulsivity, severe anxiety, suicidal ideation, or acute confusion	Stop changes; clinical contact	Caregiver/clinician/emergency support if needed	Safety endpoint; no efficacy inference
Dysphagia event	Choking, aspiration concern, new inability to swallow food, or cough with exposure	Stop that delivery method; clinician/speech review as needed	Caregiver/clinician/speech therapy	Exposure is invalid unless the percent consumed and the event context are documented
Caregiver burden collapse	Repeated missed logs, caregiver distress, or inability to maintain safety monitoring	Simplify the protocol to a minimal Tier 1	Caregiver/investigator	Missingness becomes a feasibility finding

6.7 Statistical approach and interpretability grading

The statistical plan is designed for an early-stage N-of-1 architecture rather than a randomized efficacy trial. The first analytic task is to determine whether the dataset is interpretable. Only after data density, missingness, candidate isolation, medication stability, sleep context, respiratory context, safety status, and caregiver-completion burden are characterized can trajectory interpretation proceed.

Table VI-6. Statistical and analytical plan summary.

Analysis object	Method	Decision rule	Limitation
Adherence and feasibility	Completion proportions, missingness heat maps, caregiver-burden notes	$\geq 70\%$ daily/weekly completion supports feasibility	Feasibility does not imply efficacy
Baseline characterization	Minimum interpretable baseline window and domain-specific variance estimates	Baseline accepted only if medication, sleep, respiratory, and safety context are documented	Baseline may remain unstable in a severe phenotype
Functional trajectory	Within-person time-series plots by domain	Directional signal requires repeated movement across clean or qualified cycles	No causal proof without control
Candidate-cycle comparison	APPRE cycle classification: no exposure, candidate exposure, terrain-only change, confounded change	Candidate signals require isolated or interpretable exposure windows	Candidate isolation may be impossible in high-burden periods
EEG trajectory	Spectral power, P3 amplitude/latency, artifact rates, sleep/medication annotation	Network-state signal strengthens if repeated and aligned with the functional trajectory	EEG artifacts and fatigue can dominate
Biomarker trajectory	Pre/post and longitudinal markers: BDNF, p-tau217, NfL, hs-CRP, IL-6, CMP, A1c	Concordant biomarker movement supports mechanistic plausibility	Peripheral biomarkers are non-specific
Safety	AE count, severity, stopping-rule events, and candidate holds	Safety override supersedes signal interpretation	Underreporting and caregiver fatigue risk
Sensitivity analysis	Re-run summaries excluding acute illness, falls, medication changes, and severe sleep-disruption days	The signal robust to exclusion windows is stronger	May reduce analyzable sample size
Missingness analysis	Missingness by domain, day, caregiver burden, illness, and safety context	Persistent missingness triggers simplification	Missingness may be informative rather than random

Analysis object	Method	Decision rule	Limitation
Small-series extension	Shared schema with participant-specific rules	Replication is architectural, not identical-exposure replication	Requires harmonization and local oversight

Visual analysis is primary because the first question is whether the time series is interpretable. Plots should show sleep, medication deviations, candidate exposure, safety events, respiratory status, dysphagia delivery modifications, and outcome movement on aligned timelines. Effect estimates may be calculated as standardized within-person differences from baseline, slope changes, or domain-specific percent change, where instruments support such interpretation. These thresholds are decision-support tools for signal detection, not validated responder criteria.

6.8 Missing data handling

Missing data are treated as a feasibility outcome and an interpretability constraint. Missingness is described by domain, timing, caregiver burden, safety event, sleep disruption, medication change, respiratory status, and dysphagia/delivery problems. Missingness is not automatically imputed into false precision. If a domain is missing because the participant was unsafe, fatigued, hospitalized, or the caregiver's capacity collapsed, that fact is part of the result.

6.9 Pre-registration plan

Before prospective external data collection, the protocol should freeze the APPRE rule set, candidate-evidence map, participant-specific safety logic, outcome schedule, data dictionary, adverse-event definitions, confounder matrix, stopping rules, and statistical analysis plan. Amendments should be dated and versioned before outcome interpretation.

Table VI-7. Pre-registration and open-science commitments.

Commitment	Implementation	Open-science rule	Submission-grade requirement
Preregistration	OSF registration before prospective external collection	Protocol changes are versioned before analysis	Registration records canonical protocol, outcomes, confounders, APPRE rules
Open-science archive	GitHub stores protocol, forms, dictionaries, tables, and release notes	No raw medical records	The thesis contains schema and de-identified summaries only
De-identification	RAY-001 convention in thesis	Authorization-dependent naming only	All shared tables use RAY-001 unless explicit authorization changes the policy
Data sharing	Aggregate, synthetic, or de-identified schema artifacts	Privacy-preserving; no raw clinical record dump	The data dictionary explains what can and cannot be shared
Version control	Semantic versioning for protocol, forms, APPRE rules, and thesis releases	Major releases archived	Checksums and release notes remain in the repository; the thesis remains reader-facing
Analysis freeze	Outcome definitions and interpretability rules are locked before review	Prevents post-hoc optimization	Any amendment is dated and justified before outcome interpretation
External reproducibility	Qualified teams can implement the schema without a private explanation	Same logic; not necessarily the same candidates	The protocol package includes forms, matrices, and a data dictionary

6.10 Reproducibility package

A qualified external team should be able to reproduce the architecture without relying on an informal explanation from the originator. The reproducibility package requires a canonical protocol, Layer 1 candidate-universe table, safety-filter matrix, data dictionary, adverse-event form, daily and weekly logs, APPRE rule set, confounder matrix, outcome schedule, statistical plan, and version history. Reproducibility does not require identical candidates across participants. It requires identical logic for deciding eligibility, recording exposure, annotating confounding, escalating safety, and classifying interpretability.

Chapter VII - Baseline Findings and Preliminary Dataset

Chapter VII separates confirmed baseline evidence from future outcome claims. RAY-VASC is not claiming that the protocol has already produced disease-modifying results. It is documenting the baseline from which any future trajectory must be judged and identifying the gaps that must be closed before a claim can be made. Baseline data are therefore treated as measurement infrastructure: they define severity, confounders, risk, feasibility, and the starting state of the responseome.

7.1 Confirmed baseline values and domain severity

RAY-001 begins from a severe and highly confounded state. Cognitive impairment is substantial, motor burden is clinically meaningful, sleep architecture is profoundly disrupted, and the case includes vascular, Parkinsonian, respiratory, medication, dysphagia, and fall-risk layers. A baseline this complex is scientifically valuable only if it is not simplified into a single diagnosis or a single outcome score.

Table VII-1. Confirmed baseline functional and physiological values.

Domain	Baseline value/status	Interpretation	Follow-up role	Claim boundary
Global cognition	MoCA 11/30	Severe cognitive impairment baseline	Repeated cognitive anchor; interpret only with sleep and medication context	Not a standalone disease-progression marker
Parkinsonian motor severity	MDS-UPDRS III motor 52	Substantial motor burden	Motor trajectory anchor	Sensitive to medication timing, sleep, fatigue, and examiner context
Global staging	Hoehn & Yahr Stage 3	Postural instability with preserved some ambulation	Falls and mobility risk stratification	Not PSP-specific by itself
Sleep efficiency	39.48%	Severe sleep suppression	Regenerative Loop priority and major confounder	Sleep improvement can mimic cognitive/motor improvement
Periodic limb movements	PLMI 137.66/hr	Severe sleep-fragmentation burden	Sleep-clinic target and confounder flag	Requires treatment-status tracking
Renal function	eGFR approximately 65 with documented nadir 44	Renal variability affecting candidate safety	CMP/eGFR monitoring and exclusion logic	Single value insufficient for trend claim
Metabolic status	A1c 5.8%	Prediabetes-range metabolic context	Terrain and vascular-risk tracking	Needs updated labs for the status
B12	328 pg/mL	Low-normal value relevant to the cognition/neuropathy context	Replacement/monitoring discussion under clinician oversight	Not a diagnosis by itself
Respiratory status	COPD diagnosis and nebulizer/rescue-medication context	Fatigue, oxygenation, exertion, and sleep confounders	SpO2, dyspnea, rescue-use logs	Respiratory changes can distort all functional domains
Dysphagia/falls	Soft/pureed diet, recurrent falls, head-impact concern	Safety and feasibility burden	Stopping rules, swallowing precautions, and caregiver logs	High-priority safety domain

These values create a demanding test case for RAY-VASC. They also constrain interpretation. A cognitive score cannot be interpreted without a sleep context. A gait score cannot be interpreted without medication timing and fall history. A fatigue report cannot be interpreted without a respiratory status. A biomarker result cannot be interpreted without illness, renal status, medication changes, and adherence.

7.2 Imaging and functional scan summary

The imaging baseline supports the mixed-phenotype thesis. Structural imaging identifies vascular injury and atrophy; functional dopaminergic imaging supports Parkinsonian degeneration; computed tomography contributes trauma/fall context; pending or future MRI updates are needed to refine microhemorrhage, white-matter burden, infarct burden,

and PSP-pattern structural markers. The protocol does not treat imaging as a single definitive answer. It treats imaging as one layer in a multi-domain phenotype.

Table VII-2. Imaging and scan baseline.

Date/source	Modality	Reported finding/status	RAY-VASC relevance	Follow-up requirement
Nov 2023	MRI brain	Moderate-severe chronic small-vessel white-matter ischemic changes, generalized atrophy, old bilateral thalamic infarcts, old right cerebellar infarct	Establishes vascular disconnection, thalamic relay burden, and mixed neurodegenerative/vascular context	Future MRI should quantify the burden and compare the trajectory
Aug 2025	CT/head trauma context	Post-traumatic right orbital floor deformity and fall/TBI context in the source spine	Supports traumatic exposure and safety/fall-risk confounder	Trauma history remains a confounder, not a causal diagnosis
Jan 2026	DaTSCAN	Severely reduced striatal uptake; DaTQUANT is unable to process	Supports dopaminergic degeneration but is not PSP-specific	Interpret with clinical exam and medication response
Feb 2026 planned/pending source	MRI update	Planned/source-pending update for microhemorrhage, vascular burden, and PSP review	Would refine CTE/TBI, vascular, and PSP-marker interpretation	Pending; the public thesis does not claim unavailable results
Future Tier 2	EEG	Resting-state spectral power, coherence, P3 ERP	Objective network-state marker for responseome	Requires artifact control, sleep/medication timing, and repeated sessions

The bilateral thalamic infarct finding is especially important for the thesis architecture. The thalamus functions as a relay and integration hub across attention, arousal, motor, sensory, and cognitive systems. Damage in thalamic circuits can produce disconnection-like effects that interact with PSP, VCID, sleep, and respiratory burden. RAY-VASC therefore treats thalamic burden as a network-accessibility constraint that may influence cognition, gait, sleep-wake state, and response to rehabilitation.

7.3 Laboratory baseline and gap analysis

The laboratory baseline is incomplete by design because the thesis does not pretend that the protocol already has a full biomarker panel. The correct scholarly posture is to distinguish confirmed values, clinically relevant gaps, and planned data objects. The lab panel matters because candidate safety and mechanistic interpretation depend on renal function, inflammation, metabolic status, nutritional status, and neurodegeneration markers.

Table VII-3. Laboratory baseline and gap analysis.

Category	Confirmed value/status	Needed measure	Why it matters	Protocol status
Renal function	eGFR approximately 65, with an acute nadir of 44 and later recovery noted in the source spine	Repeat CMP/eGFR, BUN/creatinine trend	Filters NAC, ALA, magnesium, high-burden supplement combinations, and medication safety	Required before broad candidate expansion
Metabolic status	A1c 5.8% historical value	Updated A1c, fasting glucose/insulin if available	Vascular risk, fatigue, inflammation, and terrain stabilization	Needed for the current baseline lock
Nutritional status	B12 328 pg/mL low-normal	B12, methylmalonic acid, folate, homocysteine	Cognitive and neuropathy confounder; correction may mimic response	Needed for interpretation
Inflammation	WBC variation noted in the source spine	hs-CRP, IL-6, TNF-alpha if feasible	Terrain burden and candidate-mechanism context	Exploratory/secondary
Neurodegeneration	Not complete in public baseline	p-tau217, NfL, where feasible	Disease-context trajectory and stratification	Exploratory; not required for feasibility
Neurotrophic marker	Not complete in public baseline	Serum/plasma BDNF with caution	Plasticity-relevant context but non-specific	Exploratory only
Sleep physiology	Sleep efficiency 39.48%, PLMI 137.66/hr	Treatment-status updates, CPAP/PLMD response, wearable sleep trend	Dominant confounder for cognition/motor/mood	High priority

Category	Confirmed value/status	Needed measure	Why it matters	Protocol status
Respiratory terrain	COPD diagnosis and rescue medication context	SpO2 trend, dyspnea scores, exacerbation log	Safety and fatigue interpretation	Tier 1 daily/weekly capture
Dysphagia/adherence	Soft/pureed diet and delivery modification are needed	Diet texture, swallow event, percent consumed, delivery vehicle	Determines whether exposure was actually received and safe	Required field in the daily exposure log

The most important baseline gap is not a single missing lab; it is the absence of a fully synchronized lab, sleep, medication, and outcome snapshot. A biomarker collected during illness, medication change, or sleep collapse has a different meaning than the same biomarker collected during a stable window. The research program needs timed baseline bundles: cognitive, motor, sleep, respiratory, adverse-event, medication, and lab context around blood draws and formal assessments.

7.4 Feasibility signal from early-phase data capture

The preliminary dataset is best understood as a readiness assessment. Before RAY-VASC can test NRRH, it must show that RAY-001 can generate enough clean observations to support interpretation. Feasibility is not secondary bureaucracy; it is the first scientific result. A severe mixed phenotype may defeat a complex protocol by making adherence impossible, increasing safety events, overwhelming the caregiver, or producing confounded observations too frequently to interpret.

Table VII-4. Feasibility signal register.

Feasibility object	Current signal	Risk	Required action	Success criterion
Daily logging	Source files contain daily/weekly packet architecture	Burden may exceed household capacity	Use minimal daily core with optional tiers	$\geq 70\%$ completion over first 90 days
Medication reconciliation	Multiple medication-frequency discrepancies exist in the source spine	Attribution error and safety risk	Lock the current medication list before the prospective cycle	Medication table frozen and dated
Sleep treatment status	Sleep study metrics severe; real-world CPAP/PLMD status unclear	Sleep confounding may dominate outcomes	Track CPAP, PLMD treatment, sedation, awakenings	Sleep context available for each outcome period
Outcome battery	Measures identified across cognition, motor, sleep, respiratory, mood, EEG, and biomarkers	Over-measurement burden	Tier measures by feasibility	Tier 1 is complete before Tier 2 expansion
Safety monitoring	Falls, dysphagia, COPD, renal variability, antiplatelet context	Safety events may require protocol holds	Use stopping rules and clinician escalation	Holds documented rather than hidden
Dysphagia/exposure validity	The delivery method may change the actual exposure	Capsule/powder/liquid differences may alter adherence and safety	Add the delivery-method field to the daily data form	Percent consumed and swallow status recorded when modified
Caregiver completion	Proxy support exists, but the burden is material	Missingness and observer fatigue	Reduce the form load if the missingness rises	Missingness map reviewed weekly
Repository readiness	Published manuscript series and thesis structure exist	Version drift and private data leakage	Maintain a de-identified thesis	Git/OSF release contains no raw medical records

Feasibility interpretation should be conservative. If the protocol cannot achieve sufficient daily/weekly completion, the correct response is simplification, not narrative compensation. If candidate changes cannot be isolated, the correct response is fewer simultaneous changes. If safety events occur, the correct response is a hold and escalation, not a claim that the event is unrelated. The preliminary dataset is successful when it teaches what can and cannot be measured.

7.5 APPRE baseline readiness rating for RAY-001

APPRE readiness is the operational judgment that determines whether the case can move from characterization into interpretable observation. A participant can be scientifically important and still not ready for candidate expansion.

RAY-001 is important precisely because the readiness threshold is high: severe sleep fragmentation, COPD, falls, dysphagia, cognitive impairment, medication complexity, and renal variability create a case where mechanistic enthusiasm must be restrained by safety and data quality.

Table VII-5. APPRE baseline readiness rating.

APPRE domain	Readiness status	Rationale	Required gate before expansion
Subject profile completeness	Partial-ready	Diagnoses and major baseline metrics are present; medication and treatment-status updates remain needed	Lock current medication/sleep/respiratory status
Safety profile	Not expansion-ready without clinician review	Falls, dysphagia, COPD, renal variability, and antiplatelet history create multiple safety gates	Clinician-reviewed safety matrix and updated labs
Outcome schedule	Tier 1 ready; Tier 2 conditional	Daily/weekly outcomes can begin; EEG/biomarkers require burden and access review	Demonstrate Tier 1 adherence first
Confounder capture	Essential but not yet proven	Sleep and medication timing may dominate the signal	Daily sleep/medication/respiratory logs before candidate expansion
Dysphagia/exposure validity	Not optional	Delivery modification changes safety and exposure interpretation	The delivery-method field is active in the daily log
Candidate universe	Mapped but not active	Layer 1 is broad; Layer 2 must remain conservative	Use eligibility/hold/exclude categories only
APPRE policy cycle	Ready for documentation, not optimization	Rules can classify cycles before any efficacy claim	Freeze the rule set and record all amendments
Small-series translation	Not yet	RAY-001 must first demonstrate interpretable-cycle feasibility	Publish the feasibility and data dictionary before replication

The baseline readiness rating defines the immediate research priority: stabilize the measurement platform. The most defensible early RAY-VASC result would not be an intervention effect. It would be proof that a high-burden mixed phenotype can be documented safely and repeatedly, with enough confounder annotation to separate clean cycles from uninterpretable cycles. That result alone would justify small-series translation because it would show that the architecture survives the stress-test case.

Chapter VIII - Discussion

8.1 Positioning within the literature

RAY-VASC sits at the intersection of several literatures that rarely meet in one design: PSP and atypical Parkinsonism, vascular cognitive impairment, post-injury network compensation, sleep and glymphatic physiology, neurotrophic mycology, psychedelic neuroplasticity, N-of-1 methodology, and open-science protocol design. The thesis is not that these fields already prove one another. The thesis is that the absence of an integrative measurement architecture prevents qualified teams from testing their convergence in complex phenotypes.

The closest conventional comparator is multidomain dementia-prevention research such as FINGER, which demonstrates that cognitive-risk intervention can be organized across diet, exercise, cognitive training, and vascular monitoring in at-risk older adults [Grade A]. RAY-VASC differs in population, scale, and purpose. It is not a prevention RCT. It addresses an already complex and clinically severe phenotype, and it asks whether daily terrain variables, candidate exposures, objective outcomes, safety events, and confounders can be converted into interpretable longitudinal data.

The stroke-recovery and neurorehabilitation literature provides a second comparator. Stroke research supports the principle that surviving networks can reorganize after injury [Grade B]. RAY-VASC borrows the logic of peri-lesional and distributed compensation but applies it cautiously to a mixed neurodegenerative/vascular phenotype. Neurodegeneration is not stroke, PSP is not a single static lesion, and vascular disconnection is not interchangeable with psychiatric network flexibility. The value of RAY-VASC is that it makes the translation testable rather than rhetorical.

Psychedelic neuroplasticity research provides a third comparator. Human neuroimaging demonstrates that psilocybin can acutely alter large-scale brain-network organization, and preclinical studies support structural plasticity mechanisms [Grade B/D]. Psychiatric trials demonstrate feasibility in regulated settings but do not establish efficacy in PSP, VCID, or mixed neurovascular disease. RAY-VASC therefore treats psilocybin as a future governed Window research object, not as an active public protocol.

8.2 Architectural contribution versus prior N-of-1 work

The RAY-VASC contribution is architectural. Prior N-of-1 work establishes that single-participant designs can be rigorous when interventions, outcomes, reporting, and analysis are transparent. RAY-VASC adds a domain-specific structure for severe mixed neurovascular neurodegeneration where candidate mechanisms, daily terrain, medication changes, safety events, dysphagia, caregiver observation, and objective measures interact.

This is where APPRE becomes central. APPRE shifts the unit of analysis from exposure to an interpretable cycle. It is not enough to know that a candidate was taken, a score changed, or a caregiver observed improvement. The architecture asks whether the cycle was clean, qualified, or uninterpretable; whether sleep was adequate; whether medication timing changed; whether respiratory status was stable; whether a fall occurred; whether dysphagia modified delivery; whether the candidate was isolated; and whether safety gates were respected.

Participant-specific eligibility remains distinct from the broader evidence map (Sections 3.6 and 6.3; full specification in Appendix B).

8.3 Limitations as structured academic analysis

The first limitation is causal inference. An N-of-1 observational architecture can generate signal, feasibility evidence, safety data, trajectory maps, and hypothesis refinement. It cannot establish broad efficacy or disease modification. Even a strong within-person trajectory would require replication, controls, and careful comparison against natural history before any therapeutic inference could be made.

The second limitation is baseline instability. RAY-001 includes severe sleep disruption, COPD, medication complexity, falls, dysphagia, renal variability, and vascular injury. These factors are not nuisance variables. They are central determinants of function. A favorable change may reflect sleep stabilization, medication timing, respiratory

improvement, caregiver support, rehabilitation, regression to the mean, or reduced acute illness rather than candidate exposure.

The third limitation is measurement burden. The more complex the architecture becomes, the greater the risk of caregiver fatigue and missingness. A thesis-grade protocol can fail if it is operationally unrealistic. RAY-VASC therefore treats burden as a scientific constraint. If forms, tests, EEG sessions, biomarker schedules, or candidate logs exceed real-world capacity, simplification is required.

The fourth limitation is evidence heterogeneity. The Layer 1 candidate universe spans Grade A evidence for some reporting standards and prevention concepts, Grade B/C evidence for selected human studies, Grade D preclinical mechanisms, and Grade E hypothesis or traditional-use context. The thesis cannot flatten this heterogeneity into a single evidence claim. It must keep each candidate and mechanism in its grade lane.

The fifth limitation is public-source asymmetry. Internal project files provide detailed case and architecture provenance, while peer-reviewed sources provide external scientific support. These source classes are not equivalent. The thesis is strongest when it explicitly separates what the project documents from what the literature establishes.

8.4 N-of-1 to small-series generalizability

The path from RAY-001 to small-series research is not a path from anecdote to certainty. It is a path from one structured responseome to multiple comparable responseomes. The small-series question is whether the same schema can be applied across related phenotypes while preserving individualized safety filters. Success would mean that external teams can use the same source hierarchy, evidence grades, outcome battery, safety matrices, confounder rules, APPRE interpretability logic, and data dictionary across participants without forcing identical interventions.

The likely small-series design is a prospective observational cohort of related mixed neurovascular-neurodegenerative phenotypes. Participants would share a minimum outcome spine: feasibility, safety, sleep, motor, cognition, respiratory status, adverse events, medication stability, dysphagia/exposure validity, where relevant, and confounder logs. Advanced tiers would include EEG and biomarkers where feasible. Candidate eligibility would remain individualized, but the logic for eligibility would be shared.

Generalizability would be supported if several participants can produce interpretable cycles, if safety filters prevent predictable errors, if missingness can be managed without destroying data quality, and if APPRE classifications are intelligible to external reviewers. Generalizability would be weakened if every case requires so much customization that the shared architecture disappears. The thesis defines replication as the reuse of the interpretability system, not the reuse of a fixed stack.

8.5 Ethical framework and public dissemination

RAY-VASC involves a vulnerable population and biologically active candidate classes; ethics, therefore, begin with conservative eligibility, clinician oversight, consent, and capacity safeguards, and explicit stopping rules. The thesis discusses mechanisms, evidence grades, safety filters, and research design, but it is not an implementation manual.

Dissemination requires de-identification and source discipline. RAY-001 is the default identifier, and the thesis presents methods, schemas, evidence logic, and de-identified summaries rather than private records.

Ethical rigor also requires preservation of exclusions, holds, adverse events, missingness, nonresponse, and uninterpretable cycles. APPRE and the open-science record treat these outcomes as part of the evidence rather than as material to be omitted.

8.6 Future directions

Future directions follow the publication ladder. Papers 5-7 should formalize the Regenerative Loop, the Neurotrophic Mycology Evidence Map, and the Three-Phase Plasticity Architecture as standalone papers. A small-series extension should test whether the RAY-VASC architecture can be replicated across related phenotypes. APPRE replication should test whether independent reviewers can classify cycles consistently. EEG refinement should test whether network-state measures are feasible and informative in this population.

Any governed Window substudy must remain separate from the public observational protocol unless lawful access, clinician governance, ethics approval, capacity safeguards, and institutional monitoring exist. That future work would not ask whether psilocybin is a general treatment for neurodegeneration. It would ask whether a tightly governed network-state perturbation can be measured safely and interpreted in relation to a preloaded, well-characterized terrain state. That distinction keeps the future research question scientific.

The most immediate future direction is not escalation. It is the completion of the measurement platform: lock baseline, reconcile medications, stabilize Tier 1 logging, resolve CPAP/PLMD treatment status, record dysphagia delivery method, finalize the data dictionary, and preregister the rule set before prospective external data collection. Only after that foundation is stable should candidate expansion or small-series replication proceed.

8.7 Integration with the seven-paper program

The published manuscript series provides the evidentiary sequence beneath this thesis. Papers 1-4 establish residual-capacity theory, the N-of-1/small-series protocol, RAY-001 baseline characterization, and APPRE; Papers 5-7 are planned extensions. Their roles are summarized in Table 1 and Appendix G.

8.8 Institutional and ethical translation

The institutional translation path proceeds from de-identified documentation to clinician, methodologist, statistician, and ethics review, then to prospective registration and small-series feasibility testing. The contribution is a reproducible measurement and decision architecture that qualified teams can inspect, contest, and replicate.

Institutional review will likely focus less on the originality of the hypothesis than on the operational controls. Reviewers will ask whether the participant population is protected, whether data are interpretable, whether candidate eligibility is conservative, whether controlled-substance boundaries are airtight, whether adverse events are captured, whether dysphagia and medication timing are documented, and whether the analysis plan prevents post-hoc claims. RAY-VASC is designed to answer those questions with tables rather than persuasion.

External review depends on sustained distinctions between architecture and intervention, hypothesis and proof, mechanistic plausibility and clinical effect, and public schema and private medical record. Those distinctions are methodological requirements rather than repeated disclaimers.

Chapter IX - Conclusion

RAY-VASC proposes a general research architecture for complex neurovascular neurodegeneration. The thesis is that residual adaptive capacity may remain measurable in surviving neural systems when biological terrain, network state, confounders, candidate inputs, safety constraints, and longitudinal outcomes are documented with enough resolution. RAY-001 forced the architecture into operational form, but the scientific claim is broader than the originating case.

The Neuropathic Repair-Remap Hypothesis provides the theoretical basis. The Regenerative Loop defines the terrain. The Three-Phase Architecture defines temporal logic. APPRE defines response-policy governance. The N-of-1 protocol defines the measurement platform. The published and planned manuscript series defines the research sequence. Together, these components form a falsifiable, evidence-graded framework for investigating residual adaptive capacity in mixed neurodegenerative and cerebrovascular phenotypes.

The conclusion of the thesis is not that the hypothesis is true. The conclusion is that the hypothesis has been translated into a testable architecture. That translation is the necessary step between intuition and evidence. Without it, residual capacity remains a story. With it, residual capacity becomes a measurable claim that can be supported, weakened, or rejected by data.

The architecture is intentionally demanding because cleaner inference requires disciplined documentation, safety filtering, evidence grading, and protocol control. Failure to produce interpretable cycles under real-world constraints would require revision or rejection; success would justify replication.

Its translational value lies in linking family-originated observation to formal protocol, N-of-1 depth to small-series replication, and open documentation to institutional review.

The path forward is structured: finalize Papers 5-7, replicate APPRE across related cases, refine EEG and biomarker methods, extend into small-series designs, and test whether the architecture can survive external review, ethical scrutiny, and real-world implementation burden. The next scientific step is collaborative investigation by qualified teams capable of separating hope from signal, signal from confounding, and mechanistic plausibility from clinical proof.

Future trials and translation. The minimum viable next step beyond this thesis is a tightly governed PSP/VCID small-series cluster (3-5 participants) run under the same N-of-1 architecture, APPRE rule-set, and responseome schema. Such a series would require prior ethics approval, formal clinician governance, biostatistical collaboration, and a locked data dictionary and open-science preregistration to ensure that feasibility, safety, and interpretability are auditable. The purpose of these early trials would be measurement validation and architecture stress-testing, not efficacy demonstration or protocol marketing. Only after repeated, well-documented small-series signals could larger-controlled studies ethically explore whether specific candidate classes warrant formal efficacy testing.

Appendices

The appendices contain the full specifications required for academic, clinical, methodological, and ethics review: evidence tables, safety filters, outcome definitions, APPRE rules, confounder handling, stopping rules, data fields, falsification criteria, source integrity, bibliography status, and N-of-1 reporting alignment.

Appendix A - Full Candidate Universe Evidence Table

Appendix A contains the full candidate-universe evidence specification. Inclusion denotes evidence-map status, not participant eligibility.

Compound/class	Class	Mechanistic rationale	Human evidence	Grade	RAY-001 eligible	Notes
Lion's Mane	Functional mycology	NGF/neurite support	Small MCI trial and preclinical data	C/D	Conditional	Standardization and tolerability required
Cordyceps	Functional mycology	Mitochondrial/respiratory fatigue rationale	Limited human data outside PSP/VCID	C/E	Conditional	COPD relevance requires clinician review
Reishi	Functional mycology	Inflammation/immune modulation	Limited direct neurodegeneration evidence	C/E	Conditional	Bleeding/immune flags
Turkey Tail	Functional mycology	Beta-glucan/gut-immune axis	Oncology/immunology is stronger than cognition	C/E	Conditional	Not a direct neuroplasticity claim
Chaga/Maitake/Shiitake	Functional mycology	Polyphenol/immune/metabolic rationale	Limited direct human neuro data	D/E	Conditional or excluded	Interaction and contamination QA needed
Citicoline	Cholinergic/membrane support	Membrane phospholipid and cholinergic support	Stroke/cognition literature is mixed	B/C	Conditional	Medication-context review
Alpha-GPC	Cholinergic support	Acetylcholine precursor	Limited clinical evidence by indication	C/E	Conditional	ACh burden review
Phosphatidylserine	Membrane support	Neuronal membrane/cognitive rationale	Older cognition trials are mixed	C	Conditional	Quality and anticoagulant context
Huperzine A	AChE inhibition	Cholinergic enhancement	Dementia trials context-dependent	B/C	Pending/excluded	Sinemet/ACh review needed
Bacopa monnieri	Cognitive Turbo/nootropic botanical	Bacoside-mediated memory and antioxidant mechanisms	RCTs and meta-analysis in cognition	A/B	Conditional	Slow-onset; standardized extract questions
NAC	Antioxidant/glutathione	Oxidative stress and glutathione support	Mixed human evidence by condition	B/C	Conditional	Bleeding and renal context
Alpha-lipoic acid	Mitochondrial antioxidant	Redox/mitochondrial support	Human evidence is stronger in metabolic/neuropathy contexts	B/C	Conditional	Glucose and renal monitoring
Curcumin	Anti-inflammatory polyphenol	NF-kB/inflammation pathway	Bioavailability-dependent human evidence	B/C	Conditional/flag	Bleeding with antiplatelet therapy
Resveratrol/quercetin	Polyphenols	Sirtuin/inflammation/vascular rationale	Mixed human evidence	C/E	Conditional	Interaction review
CoQ10/PQQ	Mitochondrial support	Mitochondrial energetics	Mixed clinical evidence	C	Conditional	Medication and renal context
NR/NMN/Urolithin A	Longevity/mitochondrial candidates	NAD/mitophagy rationale	Emerging human evidence	B/C	Pending	Renal/metabolic monitoring
Melatonin/glycine/magnesium/L-theanine/apigenin	Sleep/circadian candidates	Sleep architecture and arousal modulation	Human sleep evidence varies	A-C	Conditional	Sedation/falls context
Ashwagandha/Rhodiola/ginseng/eleuthero/schisandra	Adaptogens	Stress/fatigue terrain modulation	Mixed human evidence	B/C/E	Conditional	BP, sedation, stimulant flags

Compound/class	Class	Mechanistic rationale	Human evidence	Grade	RAY-001 eligible	Notes
Omega-3/D3+K2/B-complex/L-methylfolate/B12	Vascular/metabolic micronutrients	Inflammation and deficiency correction	Human evidence by deficiency/risk status	A-C	Conditional	Labs guide eligibility
Dihexa/9-MBC/NSI-189	Experimental future governed	Research-only neurotrophic hypotheses	No appropriate current clinical evidence for this phenotype	D/E	Excluded	Institutional/IND /IRB only

Appendix B - RAY-001 Safety Filter Matrix

Appendix B contains the full RAY-001 participant-specific safety and eligibility specification.

Candidate/class	Layer 2 status	Risk type	Safety mechanism	Monitoring required	Reinstatement condition
Lion's Mane	Conditional	Allergy/GI/standardization	Functional mushroom candidate with limited human MCI evidence	Tolerability, product QA, GI symptoms	Stable tolerance and no interaction concern
Cordyceps	Conditional/clinician-review	Respiratory/cardiovascular/fatigue attribution	COPD and exertion variables can confound perceived energy	SpO2, dyspnea, heart rate, rescue-medication use	Stable respiratory baseline and clinician review
Reishi	Hold/clinician-review	Bleeding/immune modulation	Antiplatelet history and fall risk increase concern	Bruising/bleeding log, medication review	Clinician approval and stable safety context
Turkey Tail	Conditional	GI/immune attribution	Not a direct neuroplasticity candidate	GI tolerance, infection/immune context	Stable tolerance
Chaga/Maitake/Shiitake	Hold unless QA clears	Contamination/oxalate/renal/immune	Renal variability and quality-control concerns	Renal labs, product testing	Updated renal stability and verified product QA
Citicoline	Conditional	Cholinergic/activation attribution	May affect cognition, sleep, headache, and agitation	Sleep, mood, headache, medication timing	Stable response and no destabilization
Alpha-GPC	Hold/clinician-review	Cholinergic burden	Cholinergic effects could confound cognition/sleep	Cognition, sleep, heart rate, and medication review	Clinician review and isolated cycle
Huperzine A	Excluded/pending	AChE/cholinergic interaction	Long half-life and cholinergic burden in a complex case	Not active	Formal clinician review only

Candidate/class	Layer 2 status	Risk type	Safety mechanism	Monitoring required	Reinstatement condition
Bacopa monnieri	Conditional	GI/sedation/slow-onset attribution	Slow 8-24 week onset can confound short-cycle interpretation	GI tolerance, sedation, adherence, cognitive schedule	Standardized extract and stable monitoring
NAC	Conditional/clinician-review	Bleeding/renal/GI	Antiplatelet history and renal variability require caution	Bruising, GI symptoms, CMP/eGFR	Updated labs and clinician review
Alpha-lipoic acid	Conditional	Glucose/renal/GI	Metabolic and renal context affects safety	Glucose symptoms, CMP/eGFR, GI tolerance	Stable labs and isolated cycle
Curcumin	Hold/clinician-review	Bleeding interaction	Antiplatelet/fall context makes bleeding risk relevant	Bruising/bleeding log	Clinician review
Omega-3 high-dose	Hold/clinician-review	Bleeding/falls	Antiplatelet and fall context	Bruising/bleeding log	Clinician review; conservative exposure only
Melatonin/glycine/magnesium/L-theanine/apigenin	Conditional	Sedation/falls	Sleep support can worsen morning falls or cognition	Morning sedation, falls, sleep diary	Stable dose timing and no falls signal
Ashwagandha/Rhodiola/ginseng/eleuthero/schisandra	Hold/clinician-review	BP, sedation, stimulation, thyroid/medication attribution	Adaptogens can affect arousal and cardiovascular state	BP/HR, sleep, agitation, medication review	Clinician review and one-at-a-time isolation
NR/NMN/Urolithin A	Pending	Renal/metabolic uncertainty	Longevity candidates require lab monitoring and quality control	CMP/eGFR, A1c, tolerability	Updated labs and protocol approval
Dihexa/9-MBC/NSI-189	Excluded	Experimental/regulatory	Not appropriate outside formal institutional oversight	Not active	IND/IRB/institutional governance only
Psilocybin	Not active; future governed substudy only	Legal/psychiatric/cardiovascular/capacity/safety	Controlled-substance exposure cannot be APPRE-tuned or self-directed	Institutional screening and AE oversight, if ever studied	Lawful, clinician-governed, ethics-approved protocol only

Appendix C - Outcome Measure Battery Full Specification

Appendix C expands the methodology outcome battery into an implementation-ready specification. It defines how each measure is administered, how frequently it is collected, what confounders must be co-captured, and how missingness is handled.

Domain	Instrument/measure	Administration mode	Frequency	Scoring/interpretation	Confounder co-capture	Missing-data rule
Feasibility	Daily/weekly completion rate	Caregiver/proxy log	Daily/weekly	Percent complete by field and domain	Caregiver burden, illness, hospitalization	Do not impute; report missingness pattern
Cognition	MoCA	Clinician or trained administrator	Baseline/monthly or clinician-guided	Total score and domain notes	Sleep, medication timing, fatigue, and infection	Missing if the administration conditions are invalid
Executive function	Trail Making A/B	Trained administrator	Monthly/Tier 2	Time and errors	Vision, motor impairment, fatigue, and medication timing	Mark invalid if motor/vision barrier dominates
Verbal fluency	Category/letter fluency	Trained administrator	Monthly/Tier 2	Word count/errors	Speech fatigue, sleep, and medication timing	Record reason for noncompletion
Motor function	Timed Up and Go	Caregiver/clinician with safety setup	Weekly/monthly	Time, assistance, safety note	Medication timing, falls, orthostasis, sleep	Do not collect if unsafe
Gait speed	10-meter walk	Caregiver/clinician	Weekly/monthly	Time/speed and assistance	Footwear, assistive device, medication timing	Mark context deviations
Parkinsonism/PSP	MDS-UPDRS III or PSPRS subset	Clinician or trained rater	Baseline/monthly/clinician-guided	Subscale score	Medication state, fatigue, examiner	Use the same state/time when possible
Sleep	Sleep diary, PSQI, wearable metrics	Caregiver/proxy and device	Daily/weekly/monthly	Sleep efficiency, awakenings, subjective quality	Sedatives, PLMI treatment, CPAP status	Device gaps reported separately from diary gaps
Respiratory	SpO2, mMRC/CAT, rescue-use log	Home/device and questionnaire	Daily/weekly	Oxygenation, dyspnea, rescue-use frequency	COPD flare, exertion, infection	Flag acute illness windows
Mood/behavior	PHQ-9, GAD-7, caregiver behavior note	Proxy/self if valid	Weekly/monthly	Score and qualitative change	Sleep, pain, medication change, stress	Mark proxy-only when self-report is invalid
Adverse events	WHO-style AE severity log	Caregiver/clinician	Event-driven	Grade, seriousness, relatedness unknown/possible	All concurrent changes	AE periods supersede efficacy interpretation
EEG	Resting EEG and P3 ERP	Trained technician/Tier 2	Scheduled sessions	Spectral power, P3 amplitude/latency, artifact rate	Sleep, medication timing, caffeine, fatigue	Artifact-heavy sessions marked exploratory

Domain	Instrument/measure	Administration mode	Frequency	Scoring/interpretation	Confounder co-capture	Missing-data rule
Biomarkers	BDNF, p-tau217, NfL, hs-CRP, IL-6, CMP, A1c	Laboratory	Baseline and intervals	Trend and context, not standalone proof	Illness, medication changes, fasting status	Missing intervals reported; no false interpolation
Caregiver burden	Short burden and feasibility note	Caregiver	Weekly	Burden trend and feasibility risk	Sleep, health, family stress	Missingness itself is treated as a burden signal

Appendix D - APPRE Rule-Set and Failure Mode Registry

Appendix D contains the full APPRE rule-set and failure-mode specification.

Domain	Input fields	Core rule	Output policy
Subject profile	Diagnosis, age, medications, comorbidities, caregiver capacity, swallowing status, fall risk	No candidate is considered without phenotype and capacity fit	Eligible, hold, excluded, or clinician-review status
Safety	Renal trend, antiplatelet context, BP/HR, COPD status, falls, dysphagia, sleep instability	Safety veto overrides mechanistic interest	Hold, exclude, simplify, or escalate
Exposure	Candidate, class, start/hold/continue/stop status, adherence, formulation, delivery method	Unlogged or non-isolated exposure is not interpretable	Downgrade the cycle grade or repeat after isolation
Outcome	Cognition, motor, sleep, mood, respiratory, caregiver, EEG, biomarkers	Outcome change requires a matched confounder context	Trajectory classification only after context review
Confounder	Sleep, illness, medication, falls, stress, exertion, dysphagia, respiratory change	Confounders annotate every signal	Clean, qualified, or uninterpretable cycle
Policy update	Weekly/monthly review, versioned rule set, amendment log	Policy cannot change retrospectively after seeing the desired outcomes	Continue, pause, simplify, hold, exclude, or clinician-review

Failure mode	Mechanism	Detection signal	Mitigation	Documentation object
Attribution error	Confounder mistaken for candidate effect	Change coincides with sleep, illness, medication, fall, respiratory, or caregiver shift	Downgrade interpretability and repeat only after stabilization	Confounder log; cycle grade
Safety overreach	Mechanistic interest overrides risk	Candidate conflicts with renal status, antiplatelet context, dysphagia, falls, COPD, or clinician guidance	Veto, hold, or clinician-review status	Safety matrix; AE log
Protocol creep	Too many changes in one cycle	Multiple new candidates, sleep changes, medication changes, or rehab changes occur together	Simplify to one change layer or mark cycle uninterpretable	Amendment log
Caregiver burden	Data capture exceeds household capacity	Missingness, fatigue, delayed forms, or inconsistent observations increase	Reduce form burden and preserve Tier 1 only	Feasibility log
Response-policy drift	Rules altered after outcome visibility	Unversioned amendments or retrospective interpretation changes	Freeze rule set and version all changes	Version history
Delivery-method ambiguity	Dysphagia modifies actual exposure	Capsule opened/crushed, mixed in food/liquid, or not fully consumed	Record vehicle, texture, percent consumed, and swallowing event	Exposure log; dysphagia field

Controlled-substance boundary	Rule	Rationale	Allowed documentation
Controlled substances	APPRE cannot tune, recommend, source, prepare, administer, or instruct use	Legal/ethical risk and no validated CDS/SaMD status	Published evidence, external protocol references, and governance conditions
Clinical decisions	APPRE cannot override licensed clinicians	Research artifact only	Medication timing logs, adverse events, and clinician notes
Capacity and consent	Any future substudy requires independent capacity/consent safeguards	Cognitive impairment creates vulnerability	Eligibility criteria and consent documentation only
Emergent risk	Safety events interrupt data collection	Participant safety precedes signal detection	AE log and escalation path
Public thesis boundary	The thesis discusses evidence and methods only	Prevents home-use conversion	Literature review and governance statements

Appendix E - Confounder Control Matrix

Appendix E contains the full confounder-control specification.

Confounder	Signal distortion risk	Capture method	Exclusion or flag window	Statistical/interpretive handling
Sleep disruption	Suppresses cognition, gait, mood, EEG, and caregiver-rated function	Sleep diary, wearable, PSQI, CPAP/PLMD treatment status	Same day and next day flagged; severe disruption can invalidate testing	Sensitivity analysis excluding poor-sleep days; annotate all outcomes
Medication timing change	Alter motor, cognition, alertness, orthostasis, sleep	Daily medication timing log and change notes	Affected cycle flagged	Stratify by medication state; do not compare off-state with on-state as equivalent
Falls/head injury	Acute neurologic and functional effects	Event log, head impact field, clinician review	Until clinically stable	Exclude from candidate inference or treat as a safety endpoint
Respiratory exacerbation	Hypoxia/fatigue effect on all functional measures	SpO2, dyspnea, rescue-use/nebulizer log, infection symptoms	Acute illness/flare window	Exclude or qualify cycle; analyze respiratory context separately
Infection/systemic inflammation	Sickness behavior, fatigue, biomarker distortion	Symptoms, temperature, WBC/CRP if available	Acute illness and recovery window	Annotate/invalidate cycle; avoid biomarker overinterpretation
Hydration/nutrition	Performance, orthostasis, absorption, GI tolerance	Daily intake notes and meal/timing field	Same-day flag	Context annotation and exposure-validity note
Dysphagia/delivery changes	Alters actual exposure and safety	Delivery-method field, texture, vehicle, percent consumed, cough/choke	Event day and the next day if symptomatic	Treat as an exposure modifier, not a minor note
Caregiver availability	Missingness and observer bias	Completion timestamp, burden rating, proxy identity if applicable	High-burden periods flagged	Missingness heat map; simplify packet if persistent
Rehabilitation/exertion changes	Motor/cognition changes are independent of candidate exposure	PT/speech/exertion log	Same day and next day flagged	Annotate trajectory; separate rehab-dominant cycles
Testing environment/practice	Familiarity, fatigue, distractions, time-of-day effects	Test condition note, time, assistance level	Affected test flagged	Maintain consistent conditions; interpret trend, not single value

Appendix F - Stopping Rules and Safety Escalation Matrix

Appendix F contains the full stopping rule and safety-escalation specification.

Trigger	Threshold	Immediate action	Responsible party	Research data consequence
Fall with head impact, loss of consciousness, new neurologic deficit, or unexplained acute decline	Any event	Stop research changes; seek clinical evaluation	Caregiver/clinician/emergency services as appropriate	Cycle marked safety event; candidate inference suspended
Acute respiratory distress, sustained concerning oxygenation, or COPD flare	Clinically concerning symptoms or SpO2 trend	Hold nonessential protocol tasks; follow the medical plan	Caregiver/clinician	Respiratory window excluded or qualified
Bleeding/bruising concern	New unexplained bruising, bleeding, black stool, or clinician concern	Hold bleeding-flagged candidates; clinician review	Caregiver/clinician	Safety endpoint; exposure cycle qualified/uninterpretable
Renal decline or abnormal CMP	Clinically meaningful eGFR drop, dehydration, AKI concern, abnormal creatinine/BUN	Hold renal-relevant candidates and unnecessary supplement complexity	Clinician	Candidate expansion is prohibited until stable
Severe sleep worsening or daytime somnolence	Sustained insomnia, hypersomnolence, confusion, or morning fall risk	Pause new sleep agents/inputs; review sleep plan	Caregiver/clinician	Sleep confounder flag; outcome interpretation downgraded
Behavioral destabilization	Hallucination, agitation, unsafe impulsivity, severe anxiety, suicidal ideation, or acute confusion	Stop changes; clinical contact	Caregiver/clinician/emergency support if needed	Safety endpoint; no efficacy inference
Dysphagia event	Choking, aspiration concern, new inability to swallow food, or cough with exposure	Stop that delivery method; clinician/speech review as needed	Caregiver/clinician/speech therapy	Exposure is invalid unless the percent consumed and the event context are documented
Caregiver burden collapse	Repeated missed logs, caregiver distress, or inability to maintain safety monitoring	Simplify the protocol to a minimal Tier 1	Caregiver/investigator	Missingness becomes a feasibility finding

Appendix G - Seven-Paper Series Crosswalk Table

Paper	Status	Function	Thesis dependency	Future relationship
Paper 1 - The Brain That Remains	Published/citable	Residual adaptive capacity and open-source protocol logic	Defines a theoretical object	Supports thesis foundation
Paper 2 - RAY-VASC Protocol	Published/citable	N-of-1/small-series measurement protocol	Defines study methods	Methods backbone
Paper 3 - RAY-001 Baseline	Published/citable	Originating case characterization	Defines stress-test phenotype	Case reference only
Paper 4 - APPRE	Published/citable	Response policy and confounder schema	Defines computational framework	APPRE chapter foundation
Paper 5 - Regenerative Loop	Future	Terrain model as a standalone paper	Extends Chapter IV	Future formalization
Paper 6 - Mycology Evidence Map	Future	Systematic Layer 1 candidate universe review	Extends Chapter III/Appendix A	Future evidence map
Paper 7 - Three-Phase Architecture	Future	Governed Window research design	Extends Chapter IV/VI	Future design paper

Appendix H - Data Dictionary

Field	Type	Required	Definition	Allowed values/notes
participant_id	string	yes	De-identified participant code	RAY-001 etc.
date	date	yes	Observation date	YYYY-MM-DD
sleep_quality	ordinal	yes	Self/caregiver sleep rating	0-10 or predefined scale
medication_change	boolean/text	yes	Medication deviation or change	Describe
candidate_exposure	categorical	yes	Candidate class status	none/start/continue/hold/stop
adverse_event	boolean/text	yes	Any AE or safety event	WHO grade, if applicable
motor_measure	numeric	conditional	TUG/gait/UPDRS-derived value	Instrument-specific
cognition_measure	numeric	conditional	MoCA/TMT/verbal fluency	Instrument-specific
confounder_flag	categorical	yes	Cycle interpretability issue	sleep/illness/fall/medication/etc.
interpretability_grade	categorical	yes	APPRE cycle quality	clean/qualified/uninterpretable

Additional dysphagia and exposure-validity fields required for RAY-001 and any participant with swallowing risk:

Field	Type	Required?	Definition	Use
delivery_method	Categorical	Conditional	Whole capsule/tablet, opened capsule, powder, liquid/suspension, food-mixed, not applicable	Determines whether exposure status is comparable across cycles
delivery_vehicle	Text/categorical	Conditional	Water, puree, applesauce, yogurt, meal, other vehicle	Captures dysphagia-related exposure modification
percent_consumed	Numeric/ordinal	Conditional	Estimated proportion consumed when delivery is modified	Distinguishes full exposure from partial exposure
swallow_event	Binary/severity	Conditional	Cough, choke, aspiration concern, refusal, prolonged swallow, none	Converts delivery safety into AE/confounder data
caregiver_assist	Binary/text	Conditional	Whether caregiver assistance was needed for administration/logging	Captures burden and reliability context

Appendix I - Dissertation Aims and Falsification Criteria

Aim	Question	Primary data	Falsification condition
Aim 1	Can the architecture achieve feasible data capture?	Completion, retention, and missingness	Persistent completion below the feasibility threshold
Aim 2	Can substrate priming improve terrain stability?	Sleep, respiratory, inflammatory, adherence	No terrain change despite clean adherence
Aim 3	Do objective markers correspond to function?	EEG/biomarkers/function	Markers and function remain unrelated
Aim 4	Can APPRE preserve interpretability?	Confounder and policy logs	Policy drift or uninterpretable cycles dominate
Aim 5	Can architecture translate beyond one case?	Small-series replication	Signals fail replication across related cases

Appendix J - Source Integrity and QA Register

Appendix J records publication integrity and source-QA controls for institutional review and open-science dissemination.

QA domain	Status rule	Public thesis handling
Source hierarchy	Internal vs external is clearly separated	Source types labeled
Naming policy	RAY-001 default public convention	Legal name authorization-dependent
Raw records	No raw medical records in thesis	Summaries only
Version control	Canonical version metadata retained in Git/OSF release records	Reader-facing thesis text excludes internal draft-defect history
Clinical boundary	Disclaimer once in the front matter	No repeated disclaimers

Appendix K - Source Integrity Note and Bibliographic Status Register

Appendix K is a source-integrity note for the single canonical thesis. It is not a preflight warning and does not place unverified claims into the main argument. Verified peer-reviewed and official sources appear in the main references. Verification-pending literature leads remain quarantined here and are not load-bearing.

Source group	Status in thesis	Use rule	Release posture
Peer-reviewed and official sources	Included in main references with DOI/URL where available	May support scientific claims according to evidence grade and population match	Citable
Published RAY-VASC Papers 1-4	Included as published RAY-VASC manuscript series with DOI links	May establish program architecture, protocol history, and thesis dependency	Citable project artifacts
Internal project documents	Not treated as peer-reviewed evidence	May support case provenance, architecture history, and operational details	De-identified public summary only

Appendix L - CONSORT CENT 2015 N-of-1 Checklist

CENT item	RAY-VASC implementation	Status
Rationale for N-of-1 design	A complex mixed phenotype requires within-person trajectory resolution	Specified
Eligibility criteria	Defined through case characterization and safety filters	Specified
Intervention description	Layer 1 universe and Layer 2 filter documented	Specified
Outcome measures	Primary/secondary/exploratory table provided	Specified
Sequence and timing	Pre-Load, governed Window, Sustain described	Specified
Adverse events	AE log and stopping rules specified	Specified
Analysis plan	Trajectory, sensitivity, and missing-data handling are described	Specified
Protocol changes	Version control and preregistration plan	Specified

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